Climate Weather Surface of Brazil- Hourly

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2024-05-03

First, we load the tidyverse library that we will use.

```
library(tidyverse) # for data manipulation

## ── Attaching core tidyverse packages ──────────────────────── tidyverse
2.0.0 ──
## ✔ dplyr     1.1.3     ✔ readr     2.1.4
## ✔ forcats   1.0.0     ✔ stringr   1.5.0
## ✔ ggplot2   3.4.3     ✔ tibble    3.2.1
## ✔ lubridate 1.9.3     ✔ tidyr     1.3.0
## ✔ purrr     1.0.2
## ── Conflicts ──────────────────────────────────────────
tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force
all conflicts to become errors

climate_weather <- read.csv("F:/Datasets/Climate
Weather/climate_weather.csv")
```

First, we should look at the data.

```
head(climate_weather)

##        wsnm elvt       lat      lon inme prov       date   yr mo da hr
prcp stp
## 1 BARBACENA 1169 -21.22837 -43.7677 A502   MG 2002-12-05 2002 12  5  0
0   0
## 2 BARBACENA 1169 -21.22837 -43.7677 A502   MG 2002-12-05 2002 12  5  1
0   0
## 3 BARBACENA 1169 -21.22837 -43.7677 A502   MG 2002-12-05 2002 12  5  2
0   0
## 4 BARBACENA 1169 -21.22837 -43.7677 A502   MG 2002-12-05 2002 12  5  3
0   0
## 5 BARBACENA 1169 -21.22837 -43.7677 A502   MG 2002-12-05 2002 12  5  4
0   0
## 6 BARBACENA 1169 -21.22837 -43.7677 A502   MG 2002-12-05 2002 12  5  5
0   0
##   smax smin gbrd temp dewp tmax dmax tmin dmin hmdy hmax hmin wdsp wdct
```

```
gust
## 1    0    0   NA    0    0    0    0    0    0    0    0    0    0    0
0
## 2    0    0   NA    0    0    0    0    0    0    0    0    0    0    0
0
## 3    0    0   NA    0    0    0    0    0    0    0    0    0    0    0
0
## 4    0    0   NA    0    0    0    0    0    0    0    0    0    0    0
0
## 5    0    0   NA    0    0    0    0    0    0    0    0    0    0    0
0
## 6    0    0   NA    0    0    0    0    0    0    0    0    0    0    0
0
```

```
columns_description <- read.csv("F:/Datasets/Climate
Weather/columns_description.csv")
columns_description
```

```
##    column_index                                    columns_pt.br
## 1             0                                             data
## 2             1                                             hora
## 3             2                   precipitacao total,horario (mm)
## 4             3     pressao atmosferica ao nivel da estacao (mb)
## 5             4 pressao atmosferica max. na hora ant. (aut) (mb)
## 6             5 pressao atmosferica min. na hora ant. (aut) (mb)
## 7             6                                radiation (kj/m2)
## 8             7                 temperatura do ar - bulbo seco (°c)
## 9             8                temperatura do ponto de orvalho (°c)
## 10            9       temperatura maxima na hora ant. (aut) (°c)
## 11           10       temperatura minima na hora ant. (aut) (°c)
## 12           11 temperatura orvalho max. na hora ant. (aut) (°c)
## 13           12 temperatura orvalho min. na hora ant. (aut) (°c)
## 14           13          umidade rel. max. na hora ant. (aut) (%)
## 15           14          umidade rel. min. na hora ant. (aut) (%)
## 16           15               umidade relativa do ar, horaria (%)
## 17           16               vento direcao horaria (gr) (° (gr))
## 18           17                         vento rajada maxima (m/s)
## 19           18                    vento velocidade horaria (m/s)
## 20           19                                           region
## 21           20                                            state
## 22           21                                          station
## 23           22                                     station_code
## 24           23                                         latitude
## 25           24                                        longitude
## 26           25                                           height
##                                              columns_en abbreviation
## 1                                                  date         date
## 2                                                  hour           hr
## 3                              total precipitation (mm)         prcp
## 4             atmospheric pressure at station height (mb)          stp
```

```
## 5  atmospheric pressure max. in the previous hour (mb)         smax
## 6  atmospheric pressure min. in the previous hour (mb)         smin
## 7                                    radiation (kj/m2)         gbrd
## 8                        air temperature - dry bulb (°c)         temp
## 9                             dew point temperature (°c)         dewp
## 10            max. temperature in the previous hour (°c)         tmax
## 11            min. temperature in the previous hour (°c)         tmin
## 12        dew temperature max. in the previous hour (°c)         dmax
## 13        dew temperature min. in the previous hour (°c)         dmin
## 14      relative humidity max. in the previous hour (%)         hmax
## 15      relative humidity min. in the previous hour (%)         hmin
## 16                            air relative humidity (%)         hmdy
## 17                              wind direction (° (gr))         wdct
## 18                             wind rajada maxima (m/s)         gust
## 19                                     wind speed (m/s)         wdsp
## 20                                               region
## 21                                                state         prov
## 22                                              station         wsnm
## 23                                         station_code         inme
## 24                                             latitude          lat
## 25                                            longitude          lon
## 26                                               height         elvt
##                                                           description
## 1                                                    date (YYYY-MM-DD)
## 2                                                         hour (HH:00)
## 3                       Amount of precipitation in millimetres (last hour)
## 4                              Atmospheric pressure at station level (mb)
## 5                 Maximum air pressure for the last hour in hPa to tenths
## 6                 Minimum air pressure for the last hour in hPa to tenths
## 7                                                  Solar radiation KJ/m2
## 8                              Air temperature (instant) in celsius degrees
## 9                        Dew point temperature (instant) in celsius degrees
## 10               Maximum temperature for the last hour in celsius degrees
## 11               Minimum temperature for the last hour in celsius degrees
## 12 Maximum dew point temperature for the last hour in celsius degrees
## 13 Minimum dew point temperature for the last hour in celsius degrees
## 14           Maximum relative humid temperature for the last hour in %
## 15           Minimum relative humid temperature for the last hour in %
## 16                                       Relative humid in % (instant)
## 17                              Wind direction in radius degrees (0-360)
## 18                                        Wind gust in metres per second
## 19                                       Wind speed in metres per second
## 20                                        Brazilian geopolitical regions
## 21                                                     State (Province)
## 22                      Name station (usually city location or nickname)
## 23                         Station number (INMET number) for the location
## 24                                                             Latitude
## 25                                                            Longitude
## 26                                                            Elevation
```

We summarize our data.

```
summary(climate_weather)

##      wsnm                elvt             lat              lon       
##  Length:1048575     Min.   :   9.0   Min.   :-21.228   Min.   :-46.95  
##  Class :character   1st Qu.:  38.0   1st Qu.:-20.750   1st Qu.:-43.77  
##  Mode  :character   Median : 156.0   Median :-19.988   Median :-40.74  
##                     Mean   : 414.5   Mean   :-19.004   Mean   :-41.59  
##                     3rd Qu.: 976.0   3rd Qu.:-19.357   3rd Qu.:-40.31  
##                     Max.   :1169.0   Max.   : -6.836   Max.   :-38.31  
##                                                                        
##      inme               prov               date                 yr      
##  Length:1048575     Length:1048575     Length:1048575     Min.   :2002  
##  Class :character   Class :character   Class :character   1st Qu.:2009  
##  Mode  :character   Mode  :character   Mode  :character   Median :2011  
##                                                           Mean   :2011  
##                                                           3rd Qu.:2014  
##                                                           Max.   :2016  
##                                                                         
##        mo               da              hr            prcp       
##  Min.   : 1.000   Min.   : 1.00   Min.   : 0.0   Min.   : 0.0    
##  1st Qu.: 4.000   1st Qu.: 8.00   1st Qu.: 5.0   1st Qu.: 0.0    
##  Median : 7.000   Median :16.00   Median :11.0   Median : 0.2    
##  Mean   : 6.495   Mean   :15.75   Mean   :11.5   Mean   : 1.1    
##  3rd Qu.: 9.000   3rd Qu.:23.00   3rd Qu.:17.0   3rd Qu.: 0.8    
##  Max.   :12.000   Max.   :31.00   Max.   :23.0   Max.   :96.8    
##                                                  NA's   :911403  
##       stp              smax             smin             gbrd       
##  Min.   :   0.0   Min.   :   0.0   Min.   :   0.0   Min.   :   0.0  
##  1st Qu.: 904.3   1st Qu.: 904.5   1st Qu.: 904.0   1st Qu.:  50.7  
##  Median : 996.3   Median : 996.6   Median : 996.1   Median : 861.3  
##  Mean   : 922.7   Mean   : 922.5   Mean   : 922.0   Mean   :1166.3  
##  3rd Qu.:1010.1   3rd Qu.:1010.4   3rd Qu.:1009.9   3rd Qu.:2070.1  
##  Max.   :1030.3   Max.   :1030.4   Max.   :1030.2   Max.   :7218.5  
##                                                     NA's   :443571  
##       temp            dewp             tmax            dmax      
##  Min.   : 0.00   Min.   :-10.00   Min.   : 0.00   Min.   :-9.30  
##  1st Qu.:18.90   1st Qu.: 14.70   1st Qu.:19.40   1st Qu.:15.20  
##  Median :22.40   Median : 17.60   Median :22.90   Median :18.10  
##  Mean   :21.68   Mean   : 16.52   Mean   :22.27   Mean   :16.99  
##  3rd Qu.:25.60   3rd Qu.: 20.00   3rd Qu.:26.40   3rd Qu.:20.50  
##  Max.   :43.50   Max.   : 35.00   Max.   :45.00   Max.   :35.70  
##                  NA's   :17                       NA's   :6      
##       tmin            dmin            hmdy            hmax       
##  Min.   : 0.00   Min.   :-9.60   Min.   :  0.0   Min.   :  0.00  
##  1st Qu.:18.50   1st Qu.:14.20   1st Qu.: 59.0   1st Qu.: 63.00  
##  Median :21.90   Median :17.10   Median : 77.0   Median : 80.00  
```

```
##  Mean   :21.11   Mean   :16.06   Mean   : 70.9   Mean   : 73.44
##  3rd Qu.:24.90   3rd Qu.:19.60   3rd Qu.: 90.0   3rd Qu.: 91.00
##  Max.   :42.30   Max.   :33.60   Max.   :100.0   Max.   :100.00
##                  NA's   :30
##       hmin            wdsp              wdct            gust
##  Min.   :  0.00   Min.   : 0.000   Min.   :  0.0   Min.   : 0.000
##  1st Qu.: 55.00   1st Qu.: 1.000   1st Qu.: 60.0   1st Qu.: 2.700
##  Median : 73.00   Median : 1.900   Median :121.0   Median : 4.700
##  Mean   : 68.18   Mean   : 2.175   Mean   :149.4   Mean   : 4.883
##  3rd Qu.: 87.00   3rd Qu.: 3.000   3rd Qu.:233.0   3rd Qu.: 6.700
##  Max.   :100.00   Max.   :19.100   Max.   :360.0   Max.   :25.000
##                   NA's   :23569                    NA's   :4979
```

We see many missing values in the total precipitation and radiation variable already. There are 28 variables with 1,048,575 rows. Lets glimpse into the climate_weather data further.

```
dplyr::glimpse(climate_weather)

## Rows: 1,048,575
## Columns: 28
## $ wsnm <chr> "BARBACENA", "BARBACENA", "BARBACENA", "BARBACENA",
"BARBACENA", …
## $ elvt <int> 1169, 1169, 1169, 1169, 1169, 1169, 1169, 1169, 1169, 1169,
1169,…
## $ lat  <dbl> -21.22837, -21.22837, -21.22837, -21.22837, -21.22837, -
21.22837,…
## $ lon  <dbl> -43.7677, -43.7677, -43.7677, -43.7677, -43.7677, -43.7677, -
43.7…
## $ inme <chr> "A502", "A502", "A502", "A502", "A502", "A502", "A502",
"A502", "…
## $ prov <chr> "MG", "MG", "MG", "MG", "MG", "MG", "MG", "MG", "MG", "MG",
"MG",…
## $ date <chr> "2002-12-05", "2002-12-05", "2002-12-05", "2002-12-05",
"2002-12-…
## $ yr   <int> 2002, 2002, 2002, 2002, 2002, 2002, 2002, 2002, 2002, 2002,
2002,…
## $ mo   <int> 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12,
12, 1…
## $ da   <int> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,
5, 5,…
## $ hr   <int> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17,
18,…
## $ prcp <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ stp  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ smax <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ smin <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0,…
## $ gbrd <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ temp <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ dewp <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ tmax <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ dmax <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ tmin <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ dmin <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ hmdy <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ hmax <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ hmin <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ wdsp <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ wdct <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ gust <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
```

We can skim the climate_weather data with little more details.

```
skimr::skim(climate_weather)
```

Data summary

Name	climate_weather
Number of rows	1048575
Number of columns	28
_______________________	
Column type frequency:	
character	4
numeric	24
________________________	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
wsnm	0	1	5	14	0	13	0
inme	0	1	4	4	0	13	0
prov	0	1	2	2	0	3	0
date	0	1	10	10	0	5049	0

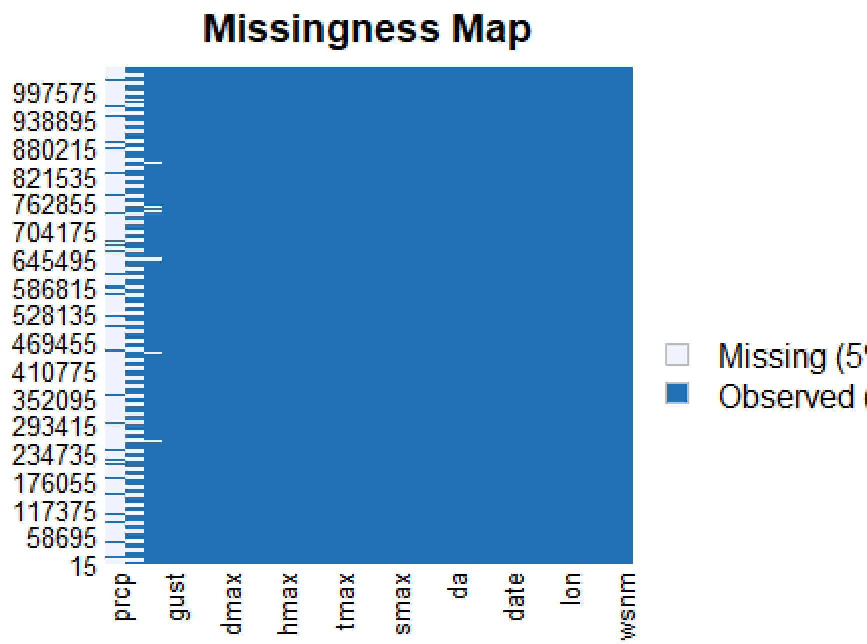
Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
elvt	0	1.00	414.53	454.43	9.00	38.00	156.00	976.00	1169.00	▇▁▁▁▂
lat	0	1.00	-19.00	3.58	-21.23	-20.75	-19.99	-19.36	-6.84	▇▁▁▁▁
lon	0	1.00	-41.59	2.37	-46.95	-43.77	-40.74	-40.31	-38.31	▂▂▁▇▁
yr	0	1.00	2010.95	3.39	2002.00	2009.00	2011.00	2014.00	2016.00	▁▂▇▇▇
mo	0	1.00	6.49	3.44	1.00	4.00	7.00	9.00	12.00	▇▆▆▆▇
da	0	1.00	15.75	8.80	1.00	8.00	16.00	23.00	31.00	▇▇▇▇▆
hr	0	1.00	11.50	6.92	0.00	5.00	11.00	17.00	23.00	▇▇▆▇▇
prcp	911403	0.13	1.11	3.07	0.00	0.00	0.20	0.80	96.80	▇▁▁▁▁
stp	0	1.00	922.68	214.34	0.00	904.30	996.30	1010.10	1030.30	▁▁▁▁▇
smax	0	1.00	922.51	215.29	0.00	904.50	996.60	1010.40	1030.40	▁▁▁▁▇
smin	0	1.00	921.97	215.19	0.00	904.00	996.10	1009.90	1030.20	▁▁▁▁▇
gbrd	443571	0.58	1166.30	1134.06	0.00	50.71	861.32	2070.12	7218.50	▇▃▁▁▁
temp	0	1.00	21.68	6.71	0.00	18.90	22.40	25.60	43.50	▁▂▇▃▁
dewp	17	1.00	16.52	5.26	-10.00	14.70	17.60	20.00	35.00	▁▁▅▇▁

skim_variable	n_missing	complete_rate	mean	sd	p0 0	p25	p50	p75	p100	hist
tmax	0	1.00	22.27	6.93	0.00	19.40	22.90	26.40	45.00	
dmax	6	1.00	16.99	5.33	-9.30	15.20	18.10	20.50	35.70	
tmin	0	1.00	21.11	6.52	0.00	18.50	21.90	24.90	42.30	
dmin	30	1.00	16.06	5.22	-9.60	14.20	17.10	19.60	33.60	
hmdy	0	1.00	70.90	23.93	0.00	59.00	77.00	90.00	100.00	
hmax	0	1.00	73.44	23.67	0.00	63.00	80.00	91.00	100.00	
hmin	0	1.00	68.18	24.17	0.00	55.00	73.00	87.00	100.00	
wdsp	23569	0.98	2.18	1.55	0.00	1.00	1.90	3.00	19.10	
wdct	0	1.00	149.38	108.70	0.00	60.00	121.00	233.00	360.00	
gust	4979	1.00	4.88	2.81	0.00	2.70	4.70	6.70	25.00	

Lets look at the ‘missingness’ map of the climate_weather data from Amelia package.

```
# missingness map
Amelia::missmap(climate_weather)
```



Lets make some other variables categorical/factor.

```
climate_weather_df <-
  climate_weather %>%
mutate(wsnm = as.factor(wsnm),
inme = as.factor(inme),prov = as.factor(prov),date = as.factor(date),dewp =
as.factor(dewp)) %>%
mutate_if(is.character, factor)
# skimming data
skimr::skim(climate_weather_df)
```

Data summary

Name	climate_weather_df
Number of rows	1048575
Number of columns	28
_______________________	
Column type frequency:	
factor	5
numeric	23
________________________	
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
wsnm	0	1	FALSE	13	BAR: 121176, ARA: 120840, ALE: 87096, SÃO: 87096
inme	0	1	FALSE	13	A50: 121176, A50: 120840, A61: 87096, A61: 87096
prov	0	1	FALSE	3	ES: 704472, MG: 266055, RJ: 78048
date	0	1	FALSE	5049	201: 288, 201: 288, 201: 288, 201: 288
dewp	17	1	FALSE	398	0: 52911, 18.: 11667, 18.: 11617, 18.: 11519

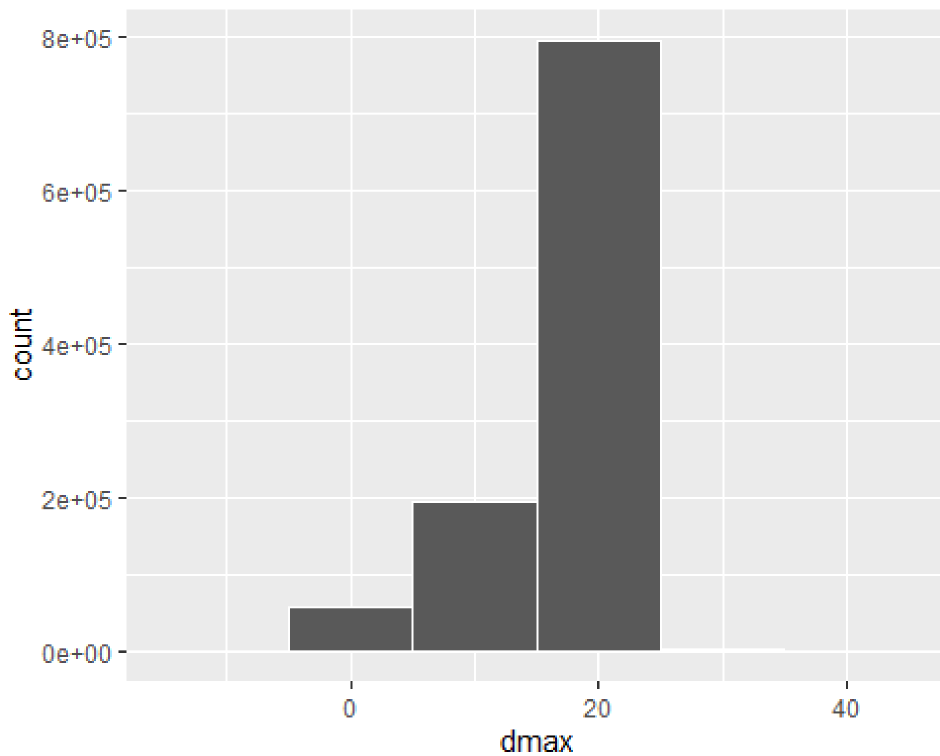
Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
elvt	0	1.00	414.53	454.43	9.00	38.00	156.00	976.00	1169.00	▇▁▁▁▂
lat	0	1.00	-19.00	3.58	-21.23	-20.75	-19.99	-19.36	-6.84	▇▁▁▁▁
lon	0	1.00	-41.59	2.37	-46.95	-43.77	-40.74	-40.31	-38.31	▂▂▁▇▂
yr	0	1.00	2010.95	3.39	2002.00	2009.00	2011.00	2014.00	2016.00	▁▂▇▇▇
mo	0	1.00	6.49	3.44	1.00	4.00	7.00	9.00	12.00	▇▆▆▆▇
da	0	1.00	15.75	8.80	1.00	8.00	16.00	23.00	31.00	▇▇▇▇▆
hr	0	1.00	11.50	6.92	0.00	5.00	11.00	17.00	23.00	▇▇▆▇▇
prcp	911403	0.13	1.11	3.07	0.00	0.00	0.20	0.80	96.80	▇▁▁▁▁
stp	0	1.00	922.68	214.34	0.00	904.30	996.30	1010.10	1030.30	▁▁▁▁▇
smax	0	1.00	922.51	215.29	0.00	904.50	996.60	1010.40	1030.40	▁▁▁▁▇

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
smin	0	1.00	921.97	215.19	0.00	904.00	996.10	1009.90	1030.20	▁▁▁▁▇
gbrd	443571	0.58	1166.30	1134.06	0.00	50.71	861.32	2070.12	7218.50	▇▃▁▁▁
temp	0	1.00	21.68	6.71	0.00	18.90	22.40	25.60	43.50	▁▃▇▃▁
tmax	0	1.00	22.27	6.93	0.00	19.40	22.90	26.40	45.00	▁▃▇▃▁
dmax	6	1.00	16.99	5.33	-9.30	15.20	18.10	20.50	35.70	▁▁▆▇▁
tmin	0	1.00	21.11	6.52	0.00	18.50	21.90	24.90	42.30	▁▃▇▃▁
dmin	30	1.00	16.06	5.22	-9.60	14.20	17.10	19.60	33.60	▁▁▆▇▁
hmdy	0	1.00	70.90	23.93	0.00	59.00	77.00	90.00	100.00	▁▁▃▆▇
hmax	0	1.00	73.44	23.67	0.00	63.00	80.00	91.00	100.00	▁▁▂▅▇
hmin	0	1.00	68.18	24.17	0.00	55.00	73.00	87.00	100.00	▁▂▅▆▇
wdsp	23569	0.98	2.18	1.55	0.00	1.00	1.90	3.00	19.10	▇▁▁▁▁
wdct	0	1.00	149.38	108.70	0.00	60.00	121.00	233.00	360.00	▇▅▃▅▅
gust	4979	1.00	4.88	2.81	0.00	2.70	4.70	6.70	25.00	▇▅▁▁▁

Look at its distribution for dmax.

```
# dmax distribution
climate_weather_df %>%
  ggplot(aes(x = dmax)) +
  geom_histogram(color = 'white', binwidth = 10)

## Warning: Removed 6 rows containing non-finite values (`stat_bin()`).
```



We will look the median of the dmax for non-missing values, and replace all missing ones with this median.

```
dmax_median <-
  climate_weather_df %>%
  summarise(median(dmax, na.rm = T))
dmax_median

##   median(dmax, na.rm = T)
## 1                    18.1
```

Median of dmax is 18.1. Now we do imputation. Note that replace function below creates a list instead of a numeric variable, so we also change it back to numeric.

```
# fill dmax variable: lazy imputation
climate_weather_df <-
  climate_weather_df %>%
  mutate(dmax = replace(dmax,is.na(dmax),dmax_median)) %>%
  mutate(dmax = unlist(dmax))
skimr::skim(climate_weather_df)
```

Data summary

Name	climate_weather_df

Number of rows	1048575
Number of columns	28
Column type frequency:	
factor	5
numeric	23
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
wsnm	0	1	FALSE	13	BAR: 121176, ARA: 120840, ALE: 87096, SÃO: 87096
inme	0	1	FALSE	13	A50: 121176, A50: 120840, A61: 87096, A61: 87096
prov	0	1	FALSE	3	ES: 704472, MG: 266055, RJ: 78048
date	0	1	FALSE	5049	201: 288, 201: 288, 201: 288, 201: 288
dewp	17	1	FALSE	398	0: 52911, 18.: 11667, 18.: 11617, 18.: 11519

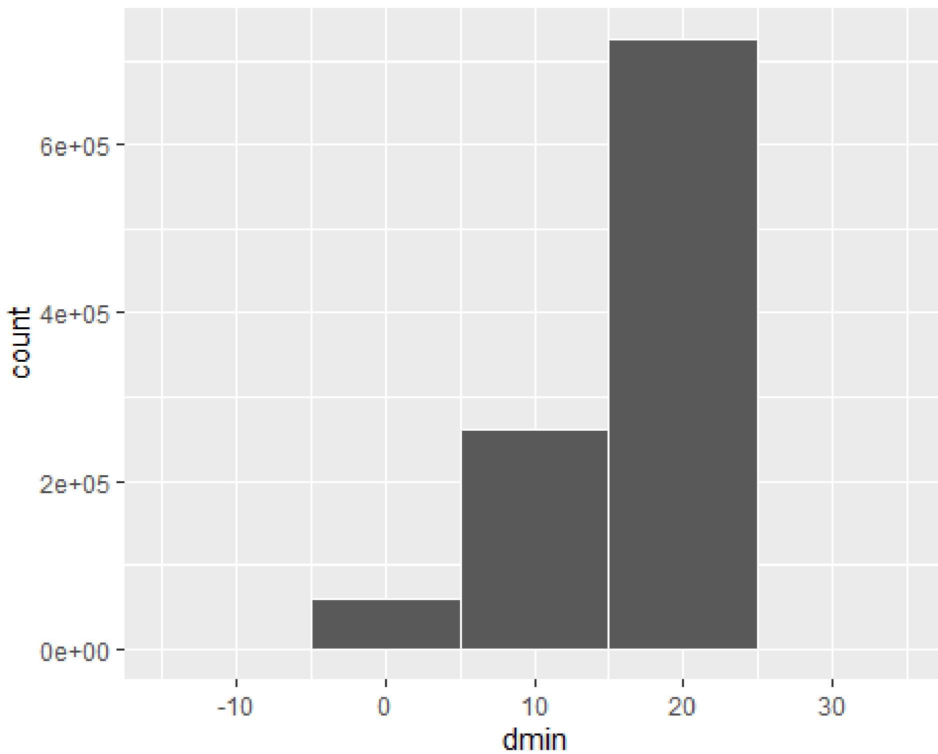
Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
elvt	0	1.00	414.53	454.43	9.00	38.00	156.00	976.00	1169.00	▇▁▁▁▂
lat	0	1.00	-19.00	3.58	-21.23	-20.75	-19.99	-19.36	-6.84	▇▁▁▁▁
lon	0	1.00	-41.59	2.37	-46.95	-43.77	-40.74	-40.31	-38.31	▂▂▁▇▂
yr	0	1.00	2010.95	3.39	2002.00	2009.00	2011.00	2014.00	2016.00	▁▂▇▇▇
mo	0	1.00	6.49	3.44	1.00	4.00	7.00	9.00	12.00	▇▆▆▆▇

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
da	0	1.00	15.75	8.80	1.00	8.00	16.00	23.00	31.00	▇▇▇▇▆
hr	0	1.00	11.50	6.92	0.00	5.00	11.00	17.00	23.00	▇▆▆▇▇
prcp	911403	0.13	1.11	3.07	0.00	0.00	0.20	0.80	96.80	▇▁▁▁▁
stp	0	1.00	922.68	214.34	0.00	904.30	996.30	1010.10	1030.30	▁▁▁▁▇
smax	0	1.00	922.51	215.29	0.00	904.50	996.60	1010.40	1030.40	▁▁▁▁▇
smin	0	1.00	921.97	215.19	0.00	904.00	996.10	1009.90	1030.20	▁▁▁▁▇
gbrd	443571	0.58	1166.30	1134.06	0.00	50.71	861.32	2070.12	7218.50	▇▃▁▁▁
temp	0	1.00	21.68	6.71	0.00	18.90	22.40	25.60	43.50	▁▃▇▃▁
tmax	0	1.00	22.27	6.93	0.00	19.40	22.90	26.40	45.00	▁▃▇▃▁
dmax	0	1.00	16.99	5.33	-9.30	15.20	18.10	20.50	35.70	▁▁▆▆▁
tmin	0	1.00	21.11	6.52	0.00	18.50	21.90	24.90	42.30	▁▃▇▃▁
dmin	30	1.00	16.06	5.22	-9.60	14.20	17.10	19.60	33.60	▁▁▆▆▁
hmdy	0	1.00	70.90	23.93	0.00	59.00	77.00	90.00	100.00	▁▁▂▆▇
hmax	0	1.00	73.44	23.67	0.00	63.00	80.00	91.00	100.00	▁▁▂▅▇
hmin	0	1.00	68.18	24.17	0.00	55.00	73.00	87.00	100.00	▁▃▅▆▇
wdsp	23569	0.98	2.18	1.55	0.00	1.00	1.90	3.00	19.10	▇▁▁▁▁
wdct	0	1.00	149.38	108.70	0.00	60.00	121.00	233.00	360.00	▇▆▅▃▅
gust	4979	1.00	4.88	2.81	0.00	2.70	4.70	6.70	25.00	▇▆▁▁▁

Let's Look at its distribution for dmin.

```
# dmin distribution
climate_weather_df %>%
  ggplot(aes(x = dmin)) +
  geom_histogram(color = 'white', binwidth = 10)

## Warning: Removed 30 rows containing non-finite values (`stat_bin()`).
```



We will look the median of the dmin for non-missing values, and replace all missing ones with this median.

```
dmin_median <-
  climate_weather_df %>%
  summarise(median(dmin, na.rm = T))
dmin_median

##   median(dmin, na.rm = T)
## 1                    17.1
```

Median of dmin is 17.1. Now we do imputation. Note that replace function below creates a lisrt instead of a numeric variable, so we also change it back to numeric.

```r
# fill dmin variable: lazy imputation
climate_weather_df <-
  climate_weather_df %>%
  mutate(dmin = replace(dmin,is.na(dmin),dmin_median)) %>%
  mutate(dmin = unlist(dmin))
skimr::skim(climate_weather_df)
```

Data summary

Name	climate_weather_df
Number of rows	1048575
Number of columns	28
_______________________	
Column type frequency:	
factor	5
numeric	23
________________________	
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
wsnm	0	1	FALSE	13	BAR: 121176, ARA: 120840, ALE: 87096, SÃO: 87096
inme	0	1	FALSE	13	A50: 121176, A50: 120840, A61: 87096, A61: 87096
prov	0	1	FALSE	3	ES: 704472, MG: 266055, RJ: 78048
date	0	1	FALSE	5049	201: 288, 201: 288, 201: 288, 201: 288
dewp	17	1	FALSE	398	0: 52911, 18.: 11667, 18.: 11617, 18.: 11519

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
elvt	0	1.00	414.53	454.43	9.00	38.00	156.00	976.00	1169.00	▇▁▁▁▂
lat	0	1.00	-19.00	3.58	-21.23	-20.75	-19.99	-19.36	-6.84	▇▁▁▁▁

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
lon	0	1.00	-41.59	2.37	-46.95	-43.77	-40.74	-40.31	-38.31	
yr	0	1.00	2010.95	3.39	2002.00	2009.00	2011.00	2014.00	2016.00	
mo	0	1.00	6.49	3.44	1.00	4.00	7.00	9.00	12.00	
da	0	1.00	15.75	8.80	1.00	8.00	16.00	23.00	31.00	
hr	0	1.00	11.50	6.92	0.00	5.00	11.00	17.00	23.00	
prcp	911403	0.13	1.11	3.07	0.00	0.00	0.20	0.80	96.80	
stp	0	1.00	922.68	214.34	0.00	904.30	996.30	1010.10	1030.30	
smax	0	1.00	922.51	215.29	0.00	904.50	996.60	1010.40	1030.40	
smin	0	1.00	921.97	215.19	0.00	904.00	996.10	1009.90	1030.20	
gbrd	443571	0.58	1166.30	1134.06	0.00	50.71	861.32	2070.12	7218.50	
temp	0	1.00	21.68	6.71	0.00	18.90	22.40	25.60	43.50	
tmax	0	1.00	22.27	6.93	0.00	19.40	22.90	26.40	45.00	
dmax	0	1.00	16.99	5.33	-9.30	15.20	18.10	20.50	35.70	
tmin	0	1.00	21.11	6.52	0.00	18.50	21.90	24.90	42.30	
dmin	0	1.00	16.06	5.22	-9.60	14.20	17.10	19.60	33.60	
hmdy	0	1.00	70.90	23.93	0.00	59.00	77.00	90.00	100.00	
hmax	0	1.00	73.44	23.67	0.00	63.00	80.00	91.00	100.00	
hmin	0	1.00	68.18	24.17	0.00	55.00	73.00	87.00	100.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
wdsp	23569	0.98	2.18	1.55	0.00	1.00	1.90	3.00	19.10	▇▁▁▁▁
wdct	0	1.00	149.38	108.70	0.00	60.00	121.00	233.00	360.00	▇▆▃▃▅
gust	4979	1.00	4.88	2.81	0.00	2.70	4.70	6.70	25.00	▇▅▁▁▁

Let's look at the distribution of dewp.

```r
# counts of each level
climate_weather_df %>%
  count(dewp)
```

```
##      dewp     n
## 1     -10     1
## 2    -9.4     1
## 3      -9     1
## 4    -8.4     1
## 5    -8.2     1
## 6    -7.4     1
## 7    -6.8     1
## 8    -6.6     2
## 9    -6.3     1
## 10   -6.1     1
## 11     -6     1
## 12   -5.2     1
## 13     -5     1
## 14   -4.9     2
## 15   -4.8     1
## 16   -4.5     1
## 17   -4.4     1
## 18   -4.3     1
## 19   -3.9     1
## 20   -3.8     4
## 21   -3.7     1
## 22   -3.5     1
## 23   -3.4     3
## 24   -3.3     2
## 25   -3.2     4
## 26   -3.1     2
## 27     -3     1
## 28   -2.9     1
## 29   -2.8     2
## 30   -2.7     2
## 31   -2.6     5
```

```
## 32  -2.5     3
## 33  -2.4     3
## 34  -2.3     4
## 35  -2.2     5
## 36  -2.1     5
## 37    -2     4
## 38  -1.9     5
## 39  -1.8     5
## 40  -1.7     4
## 41  -1.6     5
## 42  -1.5     4
## 43  -1.4     4
## 44  -1.3     2
## 45  -1.2     3
## 46  -1.1     9
## 47    -1    10
## 48  -0.9     5
## 49  -0.8     5
## 50  -0.7     7
## 51  -0.6     7
## 52  -0.5     6
## 53  -0.4    11
## 54  -0.3    10
## 55  -0.2    17
## 56  -0.1    11
## 57     0 52911
## 58   0.1    16
## 59   0.2    19
## 60   0.3    19
## 61   0.4    15
## 62   0.5    16
## 63   0.6    23
## 64   0.7    20
## 65   0.8    28
## 66   0.9    17
## 67     1    26
## 68   1.1    22
## 69   1.2    30
## 70   1.3    25
## 71   1.4    27
## 72   1.5    41
## 73   1.6    26
## 74   1.7    34
## 75   1.8    37
## 76   1.9    32
## 77     2    49
## 78   2.1    46
## 79   2.2    48
## 80   2.3    52
## 81   2.4    51
```

```
## 82   2.5    66
## 83   2.6    81
## 84   2.7    61
## 85   2.8    66
## 86   2.9    81
## 87     3    81
## 88   3.1    98
## 89   3.2    91
## 90   3.3   107
## 91   3.4   108
## 92   3.5   131
## 93   3.6   138
## 94   3.7   127
## 95   3.8   134
## 96   3.9   136
## 97     4   141
## 98   4.1   158
## 99   4.2   147
## 100  4.3   148
## 101  4.4   174
## 102  4.5   160
## 103  4.6   170
## 104  4.7   199
## 105  4.8   191
## 106  4.9   236
## 107    5   245
## 108  5.1   247
## 109  5.2   262
## 110  5.3   279
## 111  5.4   284
## 112  5.5   281
## 113  5.6   320
## 114  5.7   322
## 115  5.8   335
## 116  5.9   361
## 117    6   324
## 118  6.1   341
## 119  6.2   382
## 120  6.3   439
## 121  6.4   434
## 122  6.5   451
## 123  6.6   479
## 124  6.7   521
## 125  6.8   524
## 126  6.9   526
## 127    7   577
## 128  7.1   632
## 129  7.2   602
## 130  7.3   634
## 131  7.4   661
```

```
## 132  7.5   736
## 133  7.6   738
## 134  7.7   708
## 135  7.8   776
## 136  7.9   800
## 137    8   785
## 138  8.1   825
## 139  8.2   839
## 140  8.3   876
## 141  8.4   943
## 142  8.5   917
## 143  8.6  1001
## 144  8.7  1078
## 145  8.8  1033
## 146  8.9  1119
## 147    9  1182
## 148  9.1  1256
## 149  9.2  1246
## 150  9.3  1282
## 151  9.4  1322
## 152  9.5  1366
## 153  9.6  1447
## 154  9.7  1482
## 155  9.8  1499
## 156  9.9  1641
## 157   10  1666
## 158 10.1  1643
## 159 10.2  1700
## 160 10.3  1807
## 161 10.4  1889
## 162 10.5  1951
## 163 10.6  1954
## 164 10.7  2047
## 165 10.8  2130
## 166 10.9  2243
## 167   11  2248
## 168 11.1  2391
## 169 11.2  2382
## 170 11.3  2433
## 171 11.4  2577
## 172 11.5  2587
## 173 11.6  2807
## 174 11.7  2757
## 175 11.8  2839
## 176 11.9  2997
## 177   12  3128
## 178 12.1  3108
## 179 12.2  3298
## 180 12.3  3368
## 181 12.4  3391
```

```
## 182 12.5  3490
## 183 12.6  3548
## 184 12.7  3622
## 185 12.8  3818
## 186 12.9  3925
## 187   13  3929
## 188 13.1  4048
## 189 13.2  4254
## 190 13.3  4374
## 191 13.4  4465
## 192 13.5  4536
## 193 13.6  4527
## 194 13.7  4803
## 195 13.8  4922
## 196 13.9  5043
## 197   14  5132
## 198 14.1  5407
## 199 14.2  5364
## 200 14.3  5639
## 201 14.4  5667
## 202 14.5  5889
## 203 14.6  6227
## 204 14.7  6274
## 205 14.8  6402
## 206 14.9  6499
## 207   15  6838
## 208 15.1  7079
## 209 15.2  7137
## 210 15.3  7170
## 211 15.4  7411
## 212 15.5  7741
## 213 15.6  7788
## 214 15.7  8114
## 215 15.8  8283
## 216 15.9  8610
## 217   16  8690
## 218 16.1  8908
## 219 16.2  9162
## 220 16.3  9265
## 221 16.4  9564
## 222 16.5  9912
## 223 16.6  9922
## 224 16.7 10216
## 225 16.8 10131
## 226 16.9 10403
## 227   17 10721
## 228 17.1 10724
## 229 17.2 10896
## 230 17.3 11139
## 231 17.4 11268
```

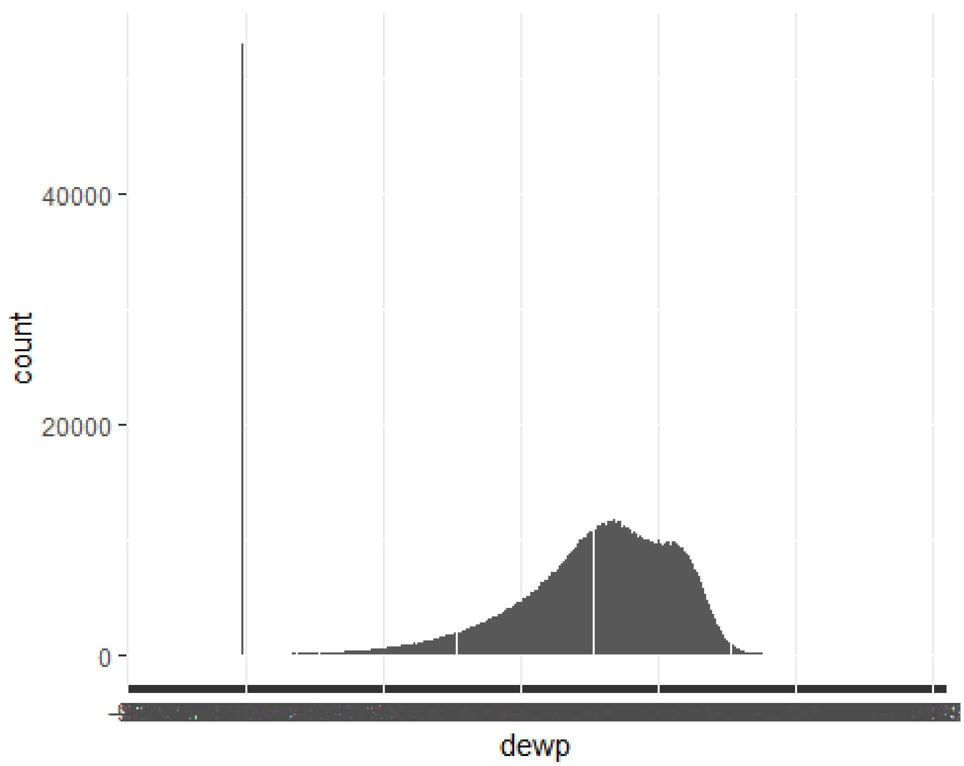
```
## 232 17.5 11367
## 233 17.6 11384
## 234 17.7 11277
## 235 17.8 11471
## 236 17.9 11481
## 237   18 11501
## 238 18.1 11667
## 239 18.2 11401
## 240 18.3 11617
## 241 18.4 11519
## 242 18.5 11084
## 243 18.6 11268
## 244 18.7 11098
## 245 18.8 10970
## 246 18.9 10879
## 247   19 10543
## 248 19.1 10738
## 249 19.2 10433
## 250 19.3 10129
## 251 19.4 10242
## 252 19.5 10214
## 253 19.6  9969
## 254 19.7  9948
## 255 19.8  9982
## 256 19.9  9843
## 257   20  9804
## 258 20.1  9699
## 259 20.2  9634
## 260 20.3  9956
## 261 20.4  9602
## 262 20.5  9486
## 263 20.6  9542
## 264 20.7  9706
## 265 20.8  9807
## 266 20.9  9474
## 267   21  9740
## 268 21.1  9778
## 269 21.2  9630
## 270 21.3  9474
## 271 21.4  9284
## 272 21.5  9219
## 273 21.6  8946
## 274 21.7  8698
## 275 21.8  8507
## 276 21.9  8298
## 277   22  7922
## 278 22.1  7297
## 279 22.2  7101
## 280 22.3  6832
## 281 22.4  6253
```

```
## 282 22.5  5817
## 283 22.6  5165
## 284 22.7  4741
## 285 22.8  4434
## 286 22.9  3859
## 287   23  3414
## 288 23.1  3065
## 289 23.2  2618
## 290 23.3  2397
## 291 23.4  1995
## 292 23.5  1723
## 293 23.6  1430
## 294 23.7  1191
## 295 23.8  1031
## 296 23.9   809
## 297   24   667
## 298 24.1   503
## 299 24.2   417
## 300 24.3   335
## 301 24.4   275
## 302 24.5   228
## 303 24.6   204
## 304 24.7   158
## 305 24.8   157
## 306 24.9   123
## 307   25   106
## 308 25.1    75
## 309 25.2    73
## 310 25.3    69
## 311 25.4    49
## 312 25.5    54
## 313 25.6    46
## 314 25.7    37
## 315 25.8    49
## 316 25.9    37
## 317   26    35
## 318 26.1    31
## 319 26.2    37
## 320 26.3    29
## 321 26.4    25
## 322 26.5    37
## 323 26.6    24
## 324 26.7    27
## 325 26.8    22
## 326 26.9    23
## 327   27    23
## 328 27.1    18
## 329 27.2    15
## 330 27.3    21
## 331 27.4    15
```

```
## 332 27.5    18
## 333 27.6    20
## 334 27.7    13
## 335 27.8    24
## 336 27.9    18
## 337   28    19
## 338 28.1    15
## 339 28.2    11
## 340 28.3    13
## 341 28.4    10
## 342 28.5     9
## 343 28.6    13
## 344 28.7     9
## 345 28.8    15
## 346 28.9    12
## 347   29    11
## 348 29.1    19
## 349 29.2    14
## 350 29.3     8
## 351 29.4     9
## 352 29.5     7
## 353 29.6     8
## 354 29.7     5
## 355 29.8     8
## 356 29.9     5
## 357   30     4
## 358 30.1     7
## 359 30.2     4
## 360 30.3     4
## 361 30.4     4
## 362 30.5     3
## 363 30.6     3
## 364 30.7     8
## 365 30.8     2
## 366 30.9     4
## 367   31     4
## 368 31.1     4
## 369 31.2     3
## 370 31.3     5
## 371 31.4     5
## 372 31.5     4
## 373 31.6     2
## 374 31.7     3
## 375 31.8     5
## 376 31.9     3
## 377   32     4
## 378 32.1     2
## 379 32.2     3
## 380 32.3     4
## 381 32.4     6
```

```
## 382 32.5      1
## 383 32.6      5
## 384 32.7      1
## 385 32.8      5
## 386 32.9      1
## 387   33      3
## 388 33.1      5
## 389 33.2      5
## 390 33.4      2
## 391 33.5      2
## 392 33.6      2
## 393 33.7      1
## 394 33.8      1
## 395 33.9      1
## 396 34.3      1
## 397 34.7      1
## 398   35      1
## 399 <NA>     17
```

```
# barplot
climate_weather_df %>%
  ggplot(aes(dewp)) +
  geom_bar()
```



The level with the highest frequency is ‘0’, and lets impute the missing values with the class that has the highest frequency.

```
# imputation
climate_weather_df <-
climate_weather_df %>%
mutate( dewp = replace(dewp,is.na(dewp),'0'))
# counts of each level after imputation
climate_weather_df %>%
count(dewp)
```

```
##     dewp    n
## 1    -10    1
## 2   -9.4    1
## 3     -9    1
## 4   -8.4    1
## 5   -8.2    1
## 6   -7.4    1
## 7   -6.8    1
## 8   -6.6    2
## 9   -6.3    1
## 10  -6.1    1
## 11    -6    1
## 12  -5.2    1
## 13    -5    1
## 14  -4.9    2
## 15  -4.8    1
## 16  -4.5    1
## 17  -4.4    1
## 18  -4.3    1
## 19  -3.9    1
## 20  -3.8    4
## 21  -3.7    1
## 22  -3.5    1
## 23  -3.4    3
## 24  -3.3    2
## 25  -3.2    4
## 26  -3.1    2
## 27    -3    1
## 28  -2.9    1
## 29  -2.8    2
## 30  -2.7    2
## 31  -2.6    5
## 32  -2.5    3
## 33  -2.4    3
## 34  -2.3    4
## 35  -2.2    5
## 36  -2.1    5
## 37    -2    4
## 38  -1.9    5
```

```
## 39  -1.8     5
## 40  -1.7     4
## 41  -1.6     5
## 42  -1.5     4
## 43  -1.4     4
## 44  -1.3     2
## 45  -1.2     3
## 46  -1.1     9
## 47    -1    10
## 48  -0.9     5
## 49  -0.8     5
## 50  -0.7     7
## 51  -0.6     7
## 52  -0.5     6
## 53  -0.4    11
## 54  -0.3    10
## 55  -0.2    17
## 56  -0.1    11
## 57     0 52928
## 58   0.1    16
## 59   0.2    19
## 60   0.3    19
## 61   0.4    15
## 62   0.5    16
## 63   0.6    23
## 64   0.7    20
## 65   0.8    28
## 66   0.9    17
## 67     1    26
## 68   1.1    22
## 69   1.2    30
## 70   1.3    25
## 71   1.4    27
## 72   1.5    41
## 73   1.6    26
## 74   1.7    34
## 75   1.8    37
## 76   1.9    32
## 77     2    49
## 78   2.1    46
## 79   2.2    48
## 80   2.3    52
## 81   2.4    51
## 82   2.5    66
## 83   2.6    81
## 84   2.7    61
## 85   2.8    66
## 86   2.9    81
## 87     3    81
## 88   3.1    98
```

```
## 89   3.2    91
## 90   3.3   107
## 91   3.4   108
## 92   3.5   131
## 93   3.6   138
## 94   3.7   127
## 95   3.8   134
## 96   3.9   136
## 97     4   141
## 98   4.1   158
## 99   4.2   147
## 100  4.3   148
## 101  4.4   174
## 102  4.5   160
## 103  4.6   170
## 104  4.7   199
## 105  4.8   191
## 106  4.9   236
## 107    5   245
## 108  5.1   247
## 109  5.2   262
## 110  5.3   279
## 111  5.4   284
## 112  5.5   281
## 113  5.6   320
## 114  5.7   322
## 115  5.8   335
## 116  5.9   361
## 117    6   324
## 118  6.1   341
## 119  6.2   382
## 120  6.3   439
## 121  6.4   434
## 122  6.5   451
## 123  6.6   479
## 124  6.7   521
## 125  6.8   524
## 126  6.9   526
## 127    7   577
## 128  7.1   632
## 129  7.2   602
## 130  7.3   634
## 131  7.4   661
## 132  7.5   736
## 133  7.6   738
## 134  7.7   708
## 135  7.8   776
## 136  7.9   800
## 137    8   785
## 138  8.1   825
```

```
## 139  8.2   839
## 140  8.3   876
## 141  8.4   943
## 142  8.5   917
## 143  8.6  1001
## 144  8.7  1078
## 145  8.8  1033
## 146  8.9  1119
## 147    9  1182
## 148  9.1  1256
## 149  9.2  1246
## 150  9.3  1282
## 151  9.4  1322
## 152  9.5  1366
## 153  9.6  1447
## 154  9.7  1482
## 155  9.8  1499
## 156  9.9  1641
## 157   10  1666
## 158 10.1  1643
## 159 10.2  1700
## 160 10.3  1807
## 161 10.4  1889
## 162 10.5  1951
## 163 10.6  1954
## 164 10.7  2047
## 165 10.8  2130
## 166 10.9  2243
## 167   11  2248
## 168 11.1  2391
## 169 11.2  2382
## 170 11.3  2433
## 171 11.4  2577
## 172 11.5  2587
## 173 11.6  2807
## 174 11.7  2757
## 175 11.8  2839
## 176 11.9  2997
## 177   12  3128
## 178 12.1  3108
## 179 12.2  3298
## 180 12.3  3368
## 181 12.4  3391
## 182 12.5  3490
## 183 12.6  3548
## 184 12.7  3622
## 185 12.8  3818
## 186 12.9  3925
## 187   13  3929
## 188 13.1  4048
```

```
## 189 13.2  4254
## 190 13.3  4374
## 191 13.4  4465
## 192 13.5  4536
## 193 13.6  4527
## 194 13.7  4803
## 195 13.8  4922
## 196 13.9  5043
## 197   14  5132
## 198 14.1  5407
## 199 14.2  5364
## 200 14.3  5639
## 201 14.4  5667
## 202 14.5  5889
## 203 14.6  6227
## 204 14.7  6274
## 205 14.8  6402
## 206 14.9  6499
## 207   15  6838
## 208 15.1  7079
## 209 15.2  7137
## 210 15.3  7170
## 211 15.4  7411
## 212 15.5  7741
## 213 15.6  7788
## 214 15.7  8114
## 215 15.8  8283
## 216 15.9  8610
## 217   16  8690
## 218 16.1  8908
## 219 16.2  9162
## 220 16.3  9265
## 221 16.4  9564
## 222 16.5  9912
## 223 16.6  9922
## 224 16.7 10216
## 225 16.8 10131
## 226 16.9 10403
## 227   17 10721
## 228 17.1 10724
## 229 17.2 10896
## 230 17.3 11139
## 231 17.4 11268
## 232 17.5 11367
## 233 17.6 11384
## 234 17.7 11277
## 235 17.8 11471
## 236 17.9 11481
## 237   18 11501
## 238 18.1 11667
```

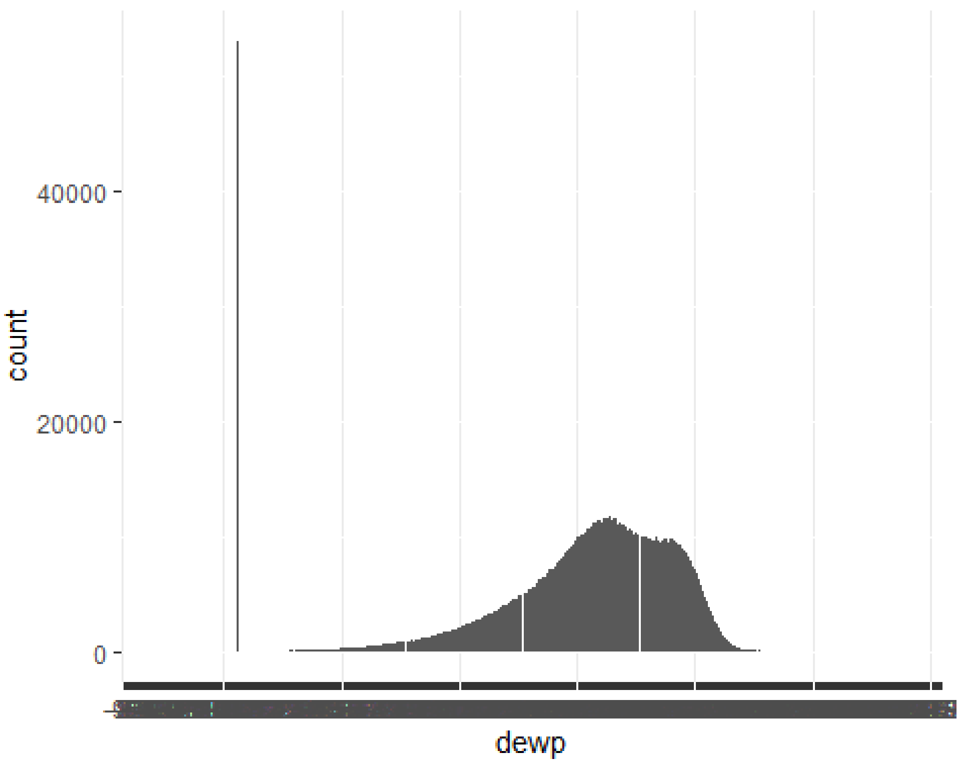
```
## 239 18.2 11401
## 240 18.3 11617
## 241 18.4 11519
## 242 18.5 11084
## 243 18.6 11268
## 244 18.7 11098
## 245 18.8 10970
## 246 18.9 10879
## 247   19 10543
## 248 19.1 10738
## 249 19.2 10433
## 250 19.3 10129
## 251 19.4 10242
## 252 19.5 10214
## 253 19.6  9969
## 254 19.7  9948
## 255 19.8  9982
## 256 19.9  9843
## 257   20  9804
## 258 20.1  9699
## 259 20.2  9634
## 260 20.3  9956
## 261 20.4  9602
## 262 20.5  9486
## 263 20.6  9542
## 264 20.7  9706
## 265 20.8  9807
## 266 20.9  9474
## 267   21  9740
## 268 21.1  9778
## 269 21.2  9630
## 270 21.3  9474
## 271 21.4  9284
## 272 21.5  9219
## 273 21.6  8946
## 274 21.7  8698
## 275 21.8  8507
## 276 21.9  8298
## 277   22  7922
## 278 22.1  7297
## 279 22.2  7101
## 280 22.3  6832
## 281 22.4  6253
## 282 22.5  5817
## 283 22.6  5165
## 284 22.7  4741
## 285 22.8  4434
## 286 22.9  3859
## 287   23  3414
## 288 23.1  3065
```

```
## 289 23.2  2618
## 290 23.3  2397
## 291 23.4  1995
## 292 23.5  1723
## 293 23.6  1430
## 294 23.7  1191
## 295 23.8  1031
## 296 23.9   809
## 297   24   667
## 298 24.1   503
## 299 24.2   417
## 300 24.3   335
## 301 24.4   275
## 302 24.5   228
## 303 24.6   204
## 304 24.7   158
## 305 24.8   157
## 306 24.9   123
## 307   25   106
## 308 25.1    75
## 309 25.2    73
## 310 25.3    69
## 311 25.4    49
## 312 25.5    54
## 313 25.6    46
## 314 25.7    37
## 315 25.8    49
## 316 25.9    37
## 317   26    35
## 318 26.1    31
## 319 26.2    37
## 320 26.3    29
## 321 26.4    25
## 322 26.5    37
## 323 26.6    24
## 324 26.7    27
## 325 26.8    22
## 326 26.9    23
## 327   27    23
## 328 27.1    18
## 329 27.2    15
## 330 27.3    21
## 331 27.4    15
## 332 27.5    18
## 333 27.6    20
## 334 27.7    13
## 335 27.8    24
## 336 27.9    18
## 337   28    19
## 338 28.1    15
```

```
## 339 28.2     11
## 340 28.3     13
## 341 28.4     10
## 342 28.5      9
## 343 28.6     13
## 344 28.7      9
## 345 28.8     15
## 346 28.9     12
## 347   29     11
## 348 29.1     19
## 349 29.2     14
## 350 29.3      8
## 351 29.4      9
## 352 29.5      7
## 353 29.6      8
## 354 29.7      5
## 355 29.8      8
## 356 29.9      5
## 357   30      4
## 358 30.1      7
## 359 30.2      4
## 360 30.3      4
## 361 30.4      4
## 362 30.5      3
## 363 30.6      3
## 364 30.7      8
## 365 30.8      2
## 366 30.9      4
## 367   31      4
## 368 31.1      4
## 369 31.2      3
## 370 31.3      5
## 371 31.4      5
## 372 31.5      4
## 373 31.6      2
## 374 31.7      3
## 375 31.8      5
## 376 31.9      3
## 377   32      4
## 378 32.1      2
## 379 32.2      3
## 380 32.3      4
## 381 32.4      6
## 382 32.5      1
## 383 32.6      5
## 384 32.7      1
## 385 32.8      5
## 386 32.9      1
## 387   33      3
## 388 33.1      5
```

```
## 389 33.2      5
## 390 33.4      2
## 391 33.5      2
## 392 33.6      2
## 393 33.7      1
## 394 33.8      1
## 395 33.9      1
## 396 34.3      1
## 397 34.7      1
## 398   35      1
```

```
# barplot
climate_weather_df %>%
ggplot(aes(dewp)) +
geom_bar()
```



Replace NA values in total precipitation(prcp), solar radiation(gbrd), wind speed(wdsp) and wind gust(gust) columns with their respective medians

```
climate_weather_df <- climate_weather_df %>% mutate(across(c(prcp, gbrd,
wdsp, gust), ~replace_na(., median(., na.rm=TRUE))))
```

```r
#skimming data
skimr::skim(climate_weather_df)
```

Data summary

Name	climate_weather_df
Number of rows	1048575
Number of columns	28
_______________________	
Column type frequency:	
factor	5
numeric	23
________________________	
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
wsnm	0	1	FALSE	13	BAR: 121176, ARA: 120840, ALE: 87096, SÃO: 87096
inme	0	1	FALSE	13	A50: 121176, A50: 120840, A61: 87096, A61: 87096
prov	0	1	FALSE	3	ES: 704472, MG: 266055, RJ: 78048
date	0	1	FALSE	5049	201: 288, 201: 288, 201: 288, 201: 288
dewp	0	1	FALSE	398	0: 52928, 18.: 11667, 18.: 11617, 18.: 11519

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
elvt	0	1	414.53	454.43	9.00	38.00	156.00	976.00	1169.00	▇▁▁▁▂
lat	0	1	-19.00	3.58	-21.23	-20.75	-19.99	-19.36	-6.84	▇▁▁▁▁
lon	0	1	-41.59	2.37	-46.95	-43.77	-40.74	-40.31	-38.31	▂▁▇▂

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
yr	0	1	2010.95	3.39	2002.00	2009.00	2011.00	2014.00	2016.00	
mo	0	1	6.49	3.44	1.00	4.00	7.00	9.00	12.00	
da	0	1	15.75	8.80	1.00	8.00	16.00	23.00	31.00	
hr	0	1	11.50	6.92	0.00	5.00	11.00	17.00	23.00	
prcp	0	1	0.32	1.15	0.00	0.20	0.20	0.20	96.80	
stp	0	1	922.68	214.34	0.00	904.30	996.30	1010.10	1030.30	
smax	0	1	922.51	215.29	0.00	904.50	996.60	1010.40	1030.40	
smin	0	1	921.97	215.19	0.00	904.00	996.10	1009.90	1030.20	
gbrd	0	1	1037.29	874.50	0.00	586.77	861.32	1173.49	7218.50	
temp	0	1	21.68	6.71	0.00	18.90	22.40	25.60	43.50	
tmax	0	1	22.27	6.93	0.00	19.40	22.90	26.40	45.00	
dmax	0	1	16.99	5.33	-9.30	15.20	18.10	20.50	35.70	
tmin	0	1	21.11	6.52	0.00	18.50	21.90	24.90	42.30	
dmin	0	1	16.06	5.22	-9.60	14.20	17.10	19.60	33.60	
hmdy	0	1	70.90	23.93	0.00	59.00	77.00	90.00	100.00	
hmax	0	1	73.44	23.67	0.00	63.00	80.00	91.00	100.00	
hmin	0	1	68.18	24.17	0.00	55.00	73.00	87.00	100.00	
wdsp	0	1	2.17	1.53	0.00	1.00	1.90	3.00	19.10	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
wdct	0	1	149.38	108.70	0.00	60.00	121.00	233.00	360.00	
gust	0	1	4.88	2.80	0.00	2.80	4.70	6.70	25.00	

Let’s check the duplicate values of all the variables in the climate weather data.

```
duplicates <- duplicated(climate_weather_df)
head(duplicates)

## [1] FALSE FALSE FALSE FALSE FALSE FALSE
```

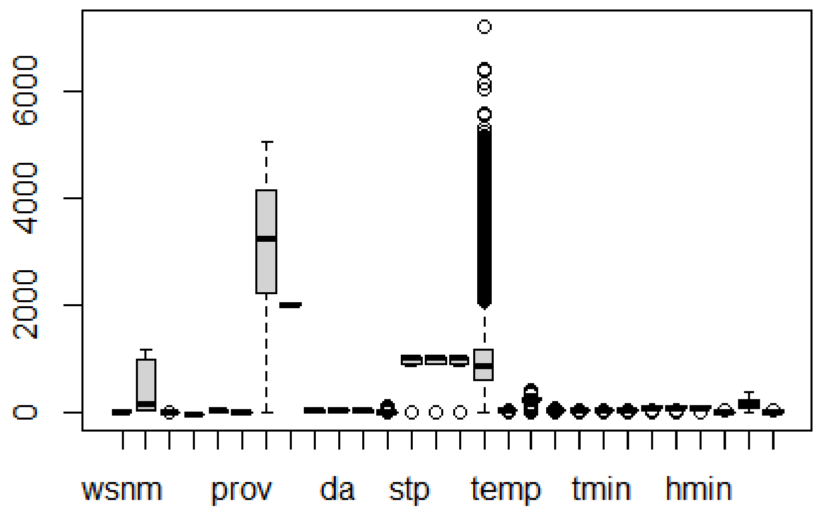
We removed all missing values and checked the duplicacy in the variables already and there are no duplicate values in any variables in the data . Hence, There are 28 variables with 1,048,575 rows. Lets glimpse into the climate_weather_df data further.

```
# glimpse
glimpse(climate_weather_df)

## Rows: 1,048,575
## Columns: 28
## $ wsnm <fct> BARBACENA, BARBACENA, BARBACENA, BARBACENA, BARBACENA,
BARBACENA,…
## $ elvt <int> 1169, 1169, 1169, 1169, 1169, 1169, 1169, 1169, 1169, 1169,
1169,…
## $ lat  <dbl> -21.22837, -21.22837, -21.22837, -21.22837, -21.22837, -
21.22837,…
## $ lon  <dbl> -43.7677, -43.7677, -43.7677, -43.7677, -43.7677, -43.7677, -
43.7…
## $ inme <fct> A502, A502, A502, A502, A502, A502, A502, A502, A502, A502,
A502,…
## $ prov <fct> MG, MG, MG, MG, MG, MG, MG, MG, MG, MG, MG, MG, MG, MG, MG,
MG, M…
## $ date <fct> 2002-12-05, 2002-12-05, 2002-12-05, 2002-12-05, 2002-12-05,
2002-…
## $ yr   <int> 2002, 2002, 2002, 2002, 2002, 2002, 2002, 2002, 2002, 2002,
2002,…
## $ mo   <int> 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12,
12, 1…
## $ da   <int> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,
5, 5,…
## $ hr   <int> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17,
18,…
## $ prcp <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ stp  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0,…
## $ smax <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ smin <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ gbrd <dbl> 861.3245, 861.3245, 861.3245, 861.3245, 861.3245, 861.3245,
861.3…
## $ temp <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ dewp <fct> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ tmax <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ dmax <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ tmin <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ dmin <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ hmdy <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ hmax <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ hmin <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ wdsp <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ wdct <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
## $ gust <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0,…
```

We can get a clear visual of the irregular data using a boxplot. There are outliers in some variable in the data.

```
boxplot(climate_weather_df)
```



Now, after performing outlier analysis in R, we replace the outliers identified by the boxplot() method with NULL values to operate over it as shown below.

```
for (x in c('stp','smax','smin','gbrd'))
{
  value = climate_weather_df[,x][climate_weather_df[,x] %in%
boxplot.stats(climate_weather_df[,x])$out]
  climate_weather_df[,x][climate_weather_df[,x] %in% value] = NA
}

#Checking whether the outliers in the above defined columns are replaced by
NULL or not
sum(is.na(climate_weather_df$stp))

## [1] 51059

sum(is.na(climate_weather_df$smax))

## [1] 51537

sum(is.na(climate_weather_df$smin))

## [1] 51538

sum(is.na(climate_weather_df$gbrd))

## [1] 153161
```

```r
as.data.frame(colSums(is.na(climate_weather_df)))
```

```
##      colSums(is.na(climate_weather_df))
## wsnm                                  0
## elvt                                  0
## lat                                   0
## lon                                   0
## inme                                  0
## prov                                  0
## date                                  0
## yr                                    0
## mo                                    0
## da                                    0
## hr                                    0
## prcp                                  0
## stp                               51059
## smax                              51537
## smin                              51538
## gbrd                             153161
## temp                                  0
## dewp                                  0
## tmax                                  0
## dmax                                  0
## tmin                                  0
## dmin                                  0
## hmdy                                  0
## hmax                                  0
## hmin                                  0
## wdsp                                  0
## wdct                                  0
## gust                                  0
```

At last, we treat the missing values by dropping the NULL values using drop_na() function from the 'tidyr' library.

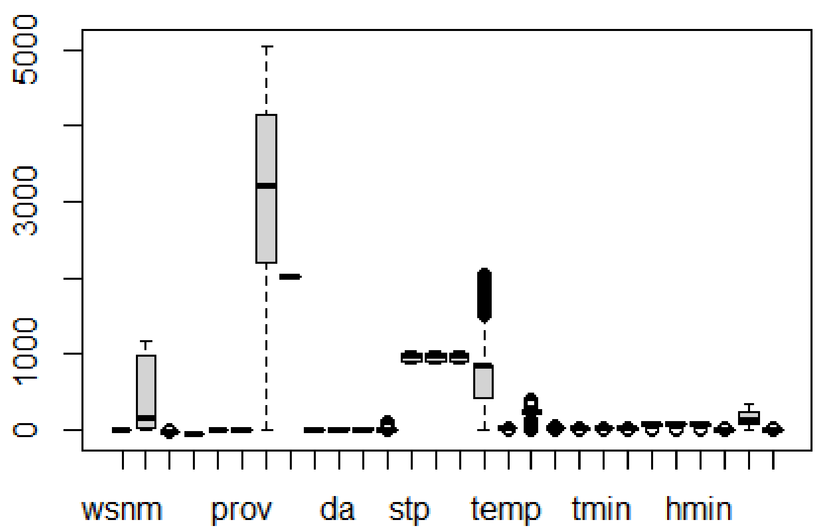
```r
#Removing the null values
climate_weather_df = drop_na(climate_weather_df)
as.data.frame(colSums(is.na(climate_weather_df)))
```

```
##      colSums(is.na(climate_weather_df))
## wsnm                                  0
## elvt                                  0
## lat                                   0
## lon                                   0
## inme                                  0
## prov                                  0
## date                                  0
## yr                                    0
## mo                                    0
```

```
## da                                        0
## hr                                        0
## prcp                                      0
## stp                                       0
## smax                                      0
## smin                                      0
## gbrd                                      0
## temp                                      0
## dewp                                      0
## tmax                                      0
## dmax                                      0
## tmin                                      0
## dmin                                      0
## hmdy                                      0
## hmax                                      0
## hmin                                      0
## wdsp                                      0
## wdct                                      0
## gust                                      0
```

After removing all the outliers, we again create boxplot.

```
boxplot(climate_weather_df)
```



Now we can clearly see that we don’t have any unclean data using summary methods.

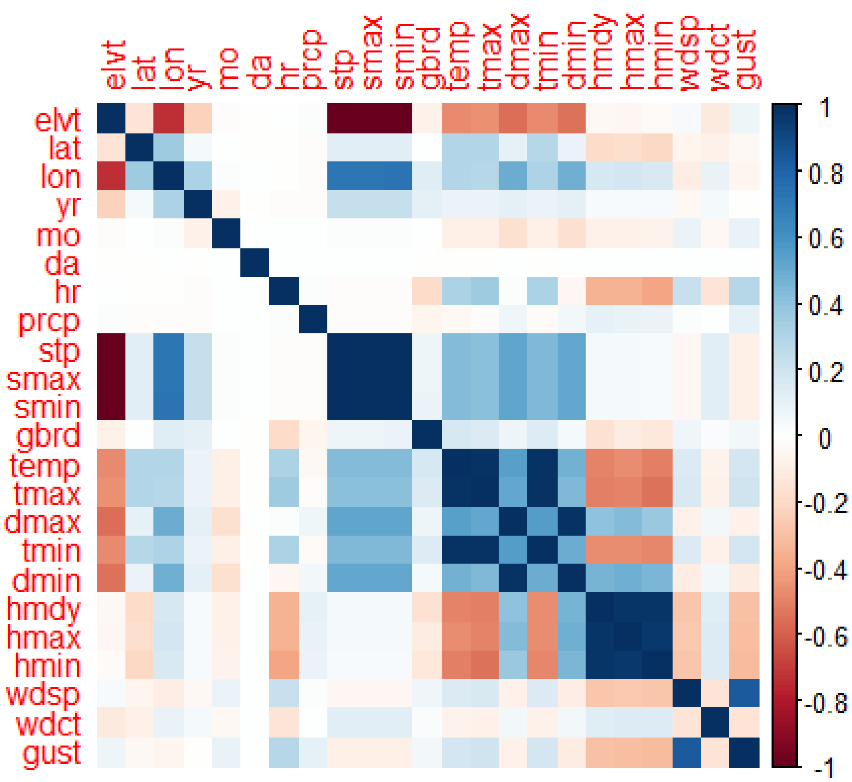
```r
summary(climate_weather_df)
```

```
##              wsnm             elvt              lat                lon
##  ARAXÁ          :102144   Min.   :   9.0   Min.   :-21.228   Min.   :-46.95
##  BARBACENA      : 95961   1st Qu.:  29.0   1st Qu.:-20.750   1st Qu.:-43.77
##  ALFREDO CHAVES: 72917   Median : 156.0   Median :-19.988   Median :-40.74
##  VITÓRIA        : 71778   Mean   : 417.8   Mean   :-19.170   Mean   :-41.67
##  SÃO MATEUS     : 71311   3rd Qu.: 976.0   3rd Qu.:-19.357   3rd Qu.:-40.31
##  LINHARES       : 70780   Max.   :1169.0   Max.   : -6.836   Max.   :-38.31
##  (Other)        :359662
##       inme           prov                  date                yr
##  A505   :102144   ES:572884   2013-07-03:   287   Min.   :2002
##  A502   : 95961   MG:221276   2015-06-25:   286   1st Qu.:2008
##  A615   : 72917   RJ: 50393   2014-07-29:   285   Median :2011
##  A612   : 71778               2012-07-17:   284   Mean   :2011
##  A616   : 71311               2012-11-09:   283   3rd Qu.:2014
##  A614   : 70780               2013-06-24:   283   Max.   :2016
##  (Other):359662               (Other)   :842845
##        mo                da              hr              prcp
##  Min.   : 1.000   Min.   : 1.00   Min.   : 0.00   Min.   : 0.0000
##  1st Qu.: 4.000   1st Qu.: 8.00   1st Qu.: 5.00   1st Qu.: 0.2000
##  Median : 7.000   Median :16.00   Median :10.00   Median : 0.2000
##  Mean   : 6.532   Mean   :15.75   Mean   :10.86   Mean   : 0.3573
##  3rd Qu.: 9.000   3rd Qu.:23.00   3rd Qu.:18.00   3rd Qu.: 0.2000
##  Max.   :12.000   Max.   :31.00   Max.   :23.00   Max.   :96.8000
##
##       stp              smax              smin              gbrd
##  Min.   : 874.0   Min.   : 877.6   Min.   : 874.0   Min.   :   0.0
##  1st Qu.: 906.2   1st Qu.: 906.4   1st Qu.: 905.9   1st Qu.: 428.6
##  Median : 998.0   Median : 998.2   Median : 997.7   Median : 861.3
##  Mean   : 969.1   Mean   : 969.3   Mean   : 968.8   Mean   : 753.8
##  3rd Qu.:1010.7   3rd Qu.:1010.8   3rd Qu.:1010.4   3rd Qu.: 861.3
##  Max.   :1030.3   Max.   :1030.4   Max.   :1030.2   Max.   :2053.6
##
##       temp             dewp              tmax             dmax
##  Min.   : 0.00   18.1    :   9826   Min.   : 0.00   Min.   :-8.00
##  1st Qu.:18.90   18      :   9780   1st Qu.:19.40   1st Qu.:15.70
##  Median :21.90   17.9    :   9739   Median :22.40   Median :18.30
##  Mean   :21.72   18.3    :   9730   Mean   :22.35   Mean   :17.76
##  3rd Qu.:24.50   17.8    :   9702   3rd Qu.:25.20   3rd Qu.:20.60
##  Max.   :40.20   18.4    :   9690   Max.   :44.30   Max.   :35.70
##                  (Other):786086
##       tmin             dmin              hmdy              hmax
##  Min.   : 0.00   Min.   :-7.60   Min.   :  0.00   Min.   :  0.00
##  1st Qu.:18.50   1st Qu.:14.90   1st Qu.: 69.00   1st Qu.: 72.00
##  Median :21.40   Median :17.50   Median : 82.00   Median : 85.00
##  Mean   :21.26   Mean   :16.95   Mean   : 78.24   Mean   : 80.41
##  3rd Qu.:24.00   3rd Qu.:19.80   3rd Qu.: 91.00   3rd Qu.: 93.00
##  Max.   :40.00   Max.   :33.60   Max.   :100.00   Max.   :100.00
##
```

```
##       hmin              wdsp              wdct             gust
##  Min.   :  0.00   Min.   : 0.000   Min.   :  0.0   Min.   : 0.000
##  1st Qu.: 65.00   1st Qu.: 1.100   1st Qu.: 73.0   1st Qu.: 2.800
##  Median : 79.00   Median : 1.900   Median :135.0   Median : 4.500
##  Mean   : 75.39   Mean   : 2.127   Mean   :158.9   Mean   : 4.801
##  3rd Qu.: 90.00   3rd Qu.: 2.900   3rd Qu.:241.0   3rd Qu.: 6.400
##  Max.   :100.00   Max.   :19.100   Max.   :360.0   Max.   :25.000
##

                                                    EDA Analysis
```

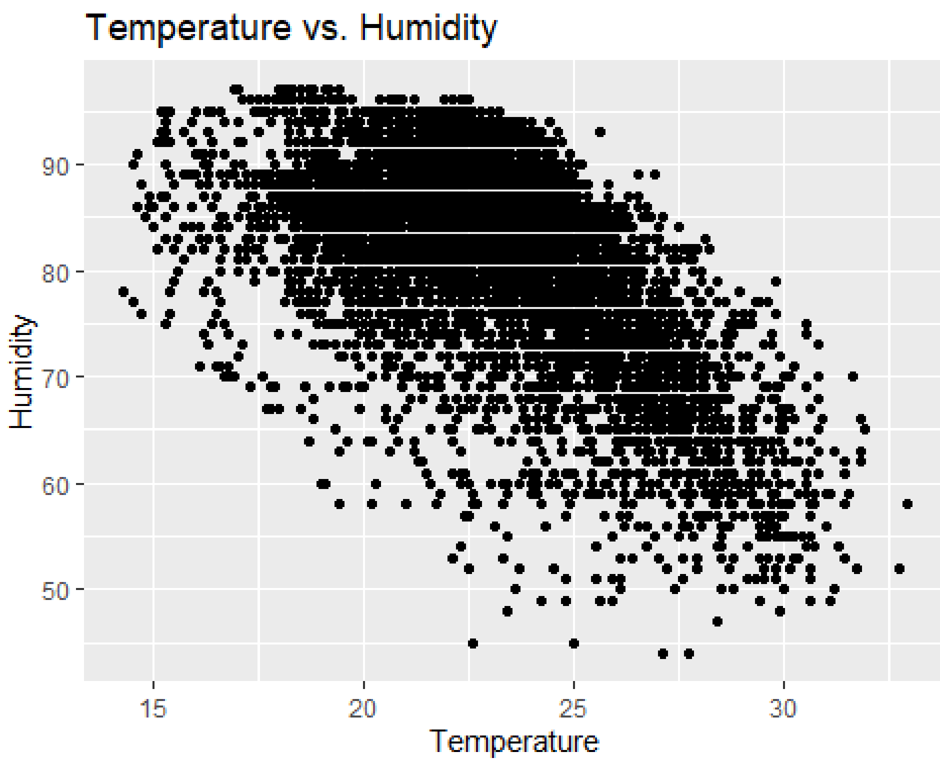
First we install and load the package 'corrplot' and create heatmap to find correlation between all the variables in the climate dataset

```r
library("corrplot")

## corrplot 0.92 loaded

# Heatmap of Correlation matrix
cor_data <- cor(select(climate_weather_df, -c(wsnm, inme, prov,dewp,date)))
print("Heatmap of Correlation matrix")

## [1] "Heatmap of Correlation matrix"

corrplot(cor_data, method="color")
```



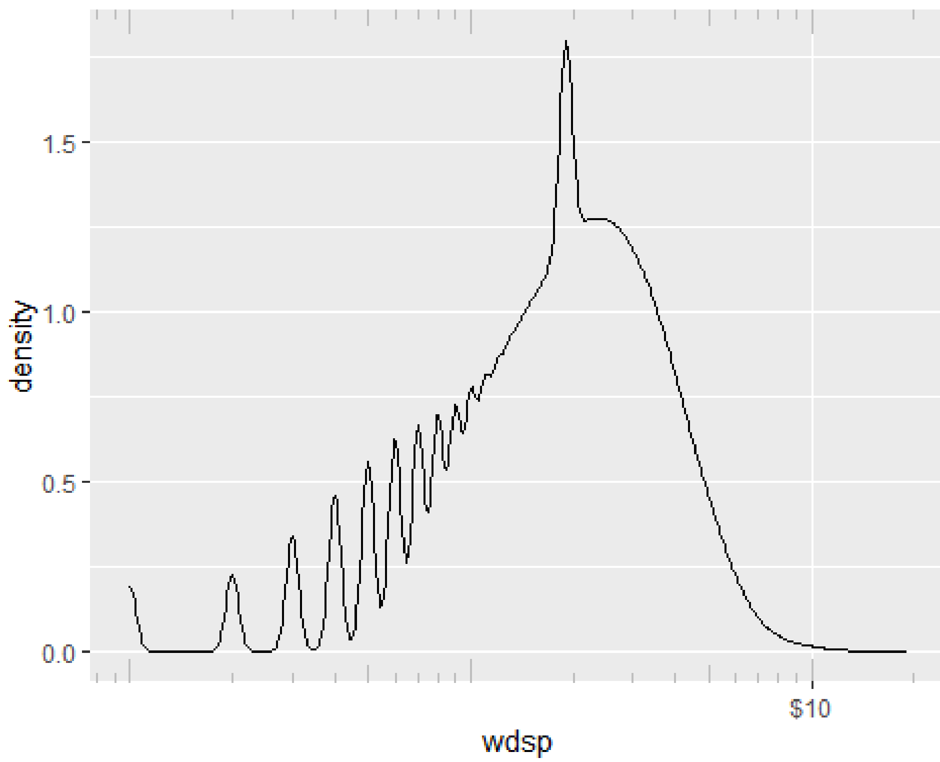
We create Scatter plot of temperature vs. humidity

```
# Scatter plot of temperature vs. humidity
df <- climate_weather_df %>% filter(yr == 2012 & wsnm=="LINHARES")
ggplot(df, aes(x = temp, y = hmdy)) +
  geom_point() +
  labs(title = "Temperature vs. Humidity", x = "Temperature", y = "Humidity")
```



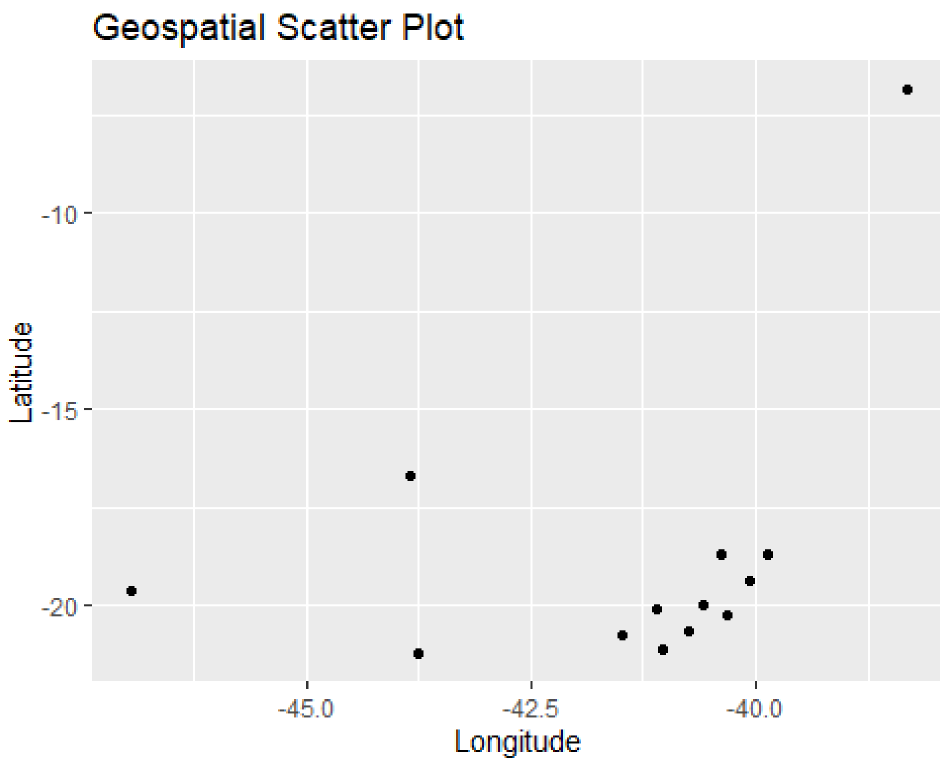
We create a density plot of wind speed which is a representation of the distribution of a wind speed variable

```
ggplot(climate_weather_df, aes(x = wdsp)) +
  geom_density() +
  scale_x_log10(breaks=c(10, 100, 1000, 10000, 100000, 1000000),
labels=scales::dollar)+
  annotation_logticks(sides="bt",color = "gray")

## Warning: Transformation introduced infinite values in continuous x-axis

## Warning: Removed 9783 rows containing non-finite values
(`stat_density()`).
```



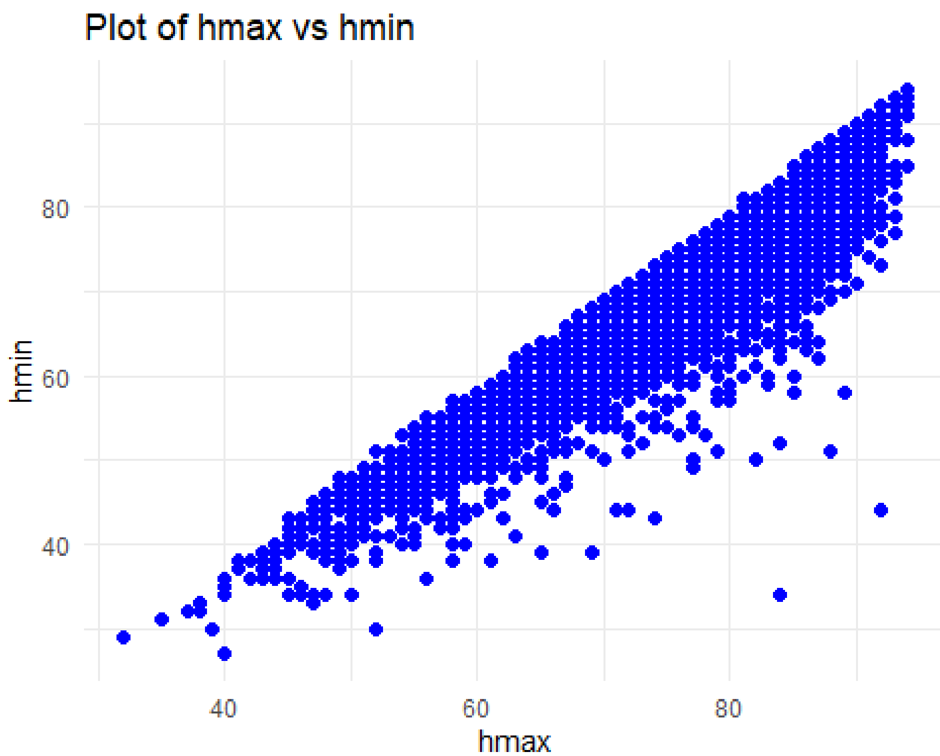
We create a scatter plot for ‘lon’ and ‘lat’ variables that present the relationship between longitude and latitute variables in a data-set

```
# creating a scatter plot for 'lon' and 'lat' variables
ggplot(climate_weather_df, aes(x = lon, y = lat)) +
  geom_point() +
  labs(title = "Geospatial Scatter Plot", x = "Longitude", y = "Latitude")
```



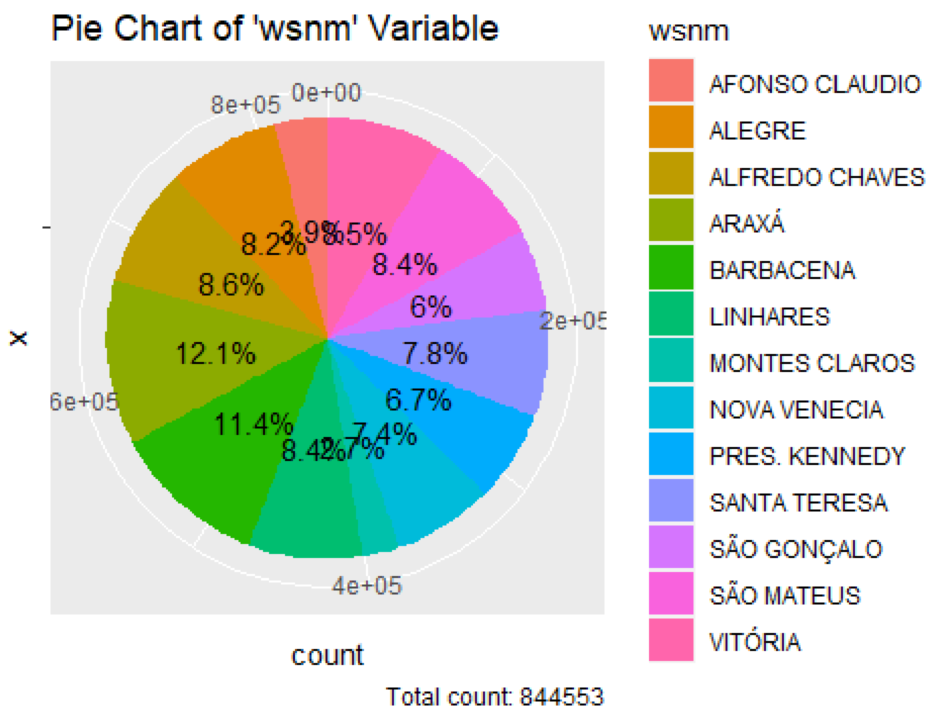
We create a scatter plot for ‘hmax’ and ‘hmin’ variables that present the relationship between maximum and minimum humidity variables in a data-set

```
library(ggplot2)

# Create a scatter plot
df <- climate_weather_df %>% filter(yr == 2010 & wsnm=="ALFREDO CHAVES")
ggplot(df, aes(x = hmax, y = hmin)) +
  geom_point(color = "blue", size = 2) +
  labs(x = "hmax", y = "hmin", title = "Plot of hmax vs hmin") +
  theme_minimal()
```



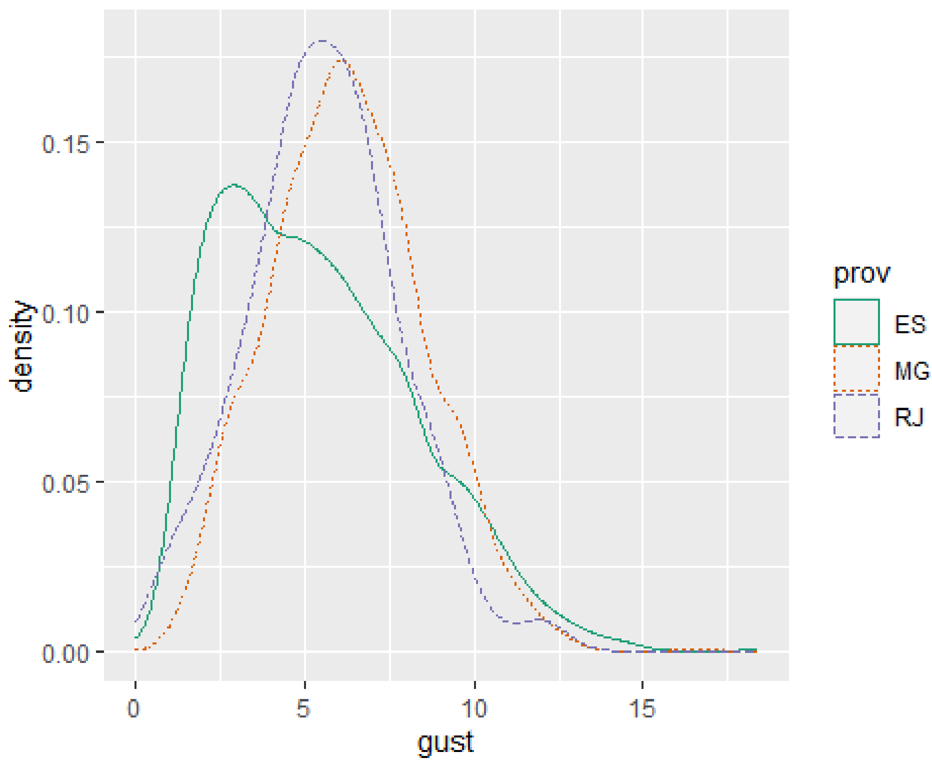
We create a Pie Chart of 'wsnm' Variable which helps organize and show data as a percentage of a whole

```
# Create a data frame with the counts of each category in 'wsnm'
wsnm_counts <- table(climate_weather_df$wsnm)
wsnm_counts_df <- data.frame(wsnm = names(wsnm_counts), count =
as.numeric(wsnm_counts))

# Calculate percentages
total <- sum(wsnm_counts_df$count)
wsnm_counts_df$percentage <- wsnm_counts_df$count / total * 100

# Create a pie chart
pie_chart <- ggplot(wsnm_counts_df, aes(x = "", y = count, fill = wsnm)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  labs(title = "Pie Chart of 'wsnm' Variable",
       caption = paste0("Total count: ", total)) +  # Add total count to the
caption
  geom_text(aes(label = paste0(round(percentage, 1), "%")), position =
position_stack(vjust = 0.5))  # Add percentages as labels

# Display the pie chart
print(pie_chart)
```



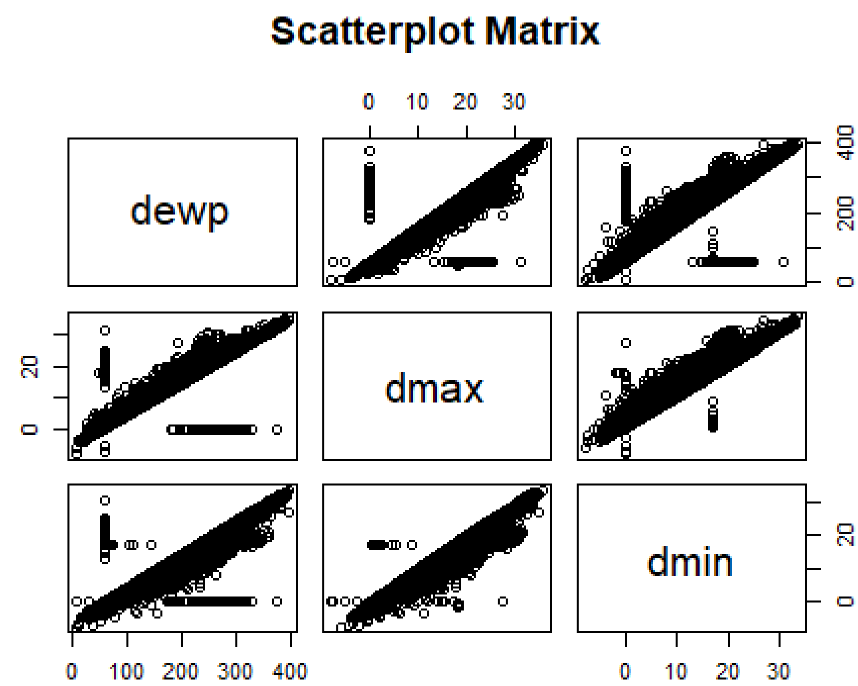
We create a density plot of wind gust by province which is a representation of the distribution of a wind gust variable

```
climate_data = subset(climate_weather_df,prov %in% c("ES","MG","RJ"))
df <- climate_data %>% filter(yr == 2011 &  mo == 09)
ggplot(df, aes(x = gust,color=prov,linetype=prov)) +
  geom_density() + scale_color_brewer(palette="Dark2")
```



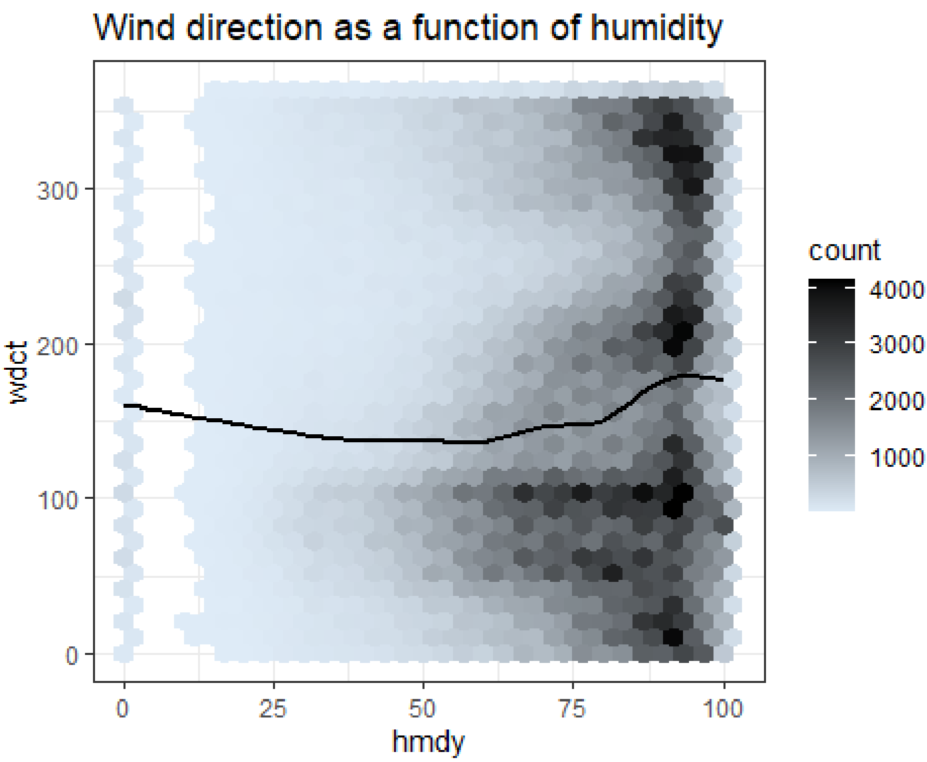
We create a scatterplot matrix between dewp, dmax, dmin variables

```
# One variable with 2 others
# and total 3 variables.
pairs(~dewp + dmax + dmin, data = climate_weather_df,
    main = "Scatterplot Matrix")
```



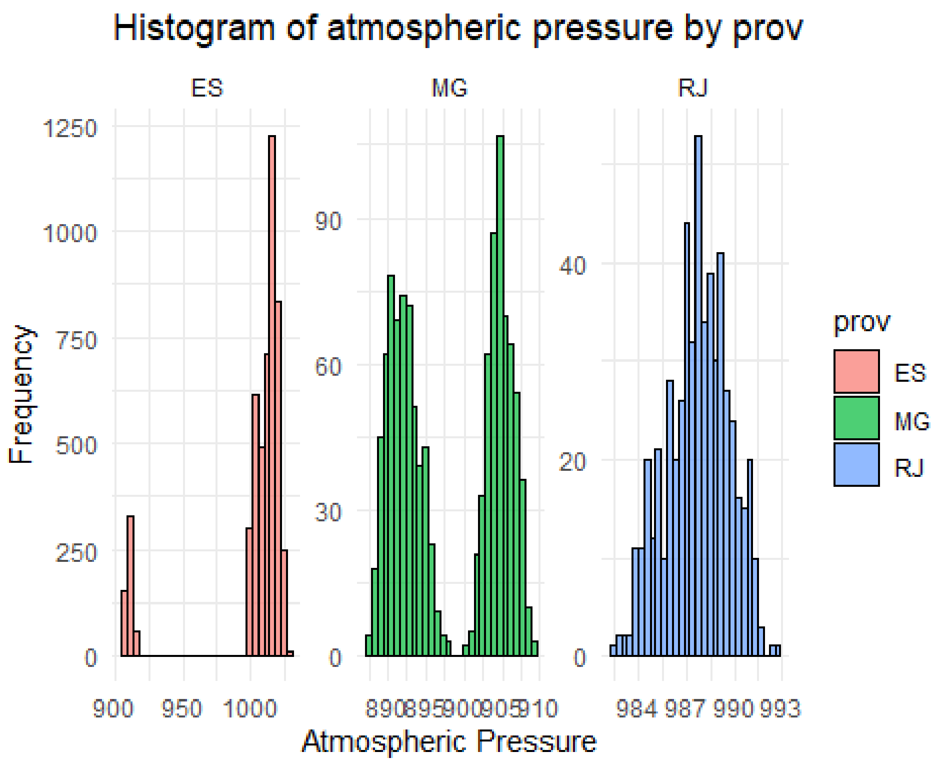
We create a HexBinPlot of hmdy vs wdct

```
library(WVPlots)

## Loading required package: wrapr

##
## Attaching package: 'wrapr'

## The following object is masked from 'package:dplyr':
##
##     coalesce

## The following objects are masked from 'package:tidyr':
##
##     pack, unpack

## The following object is masked from 'package:tibble':
##
##     view

HexBinPlot(climate_weather_df,"hmdy","wdct","Wind direction as a function of humidity")+
  geom_smooth(color="black",se=FALSE)

## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



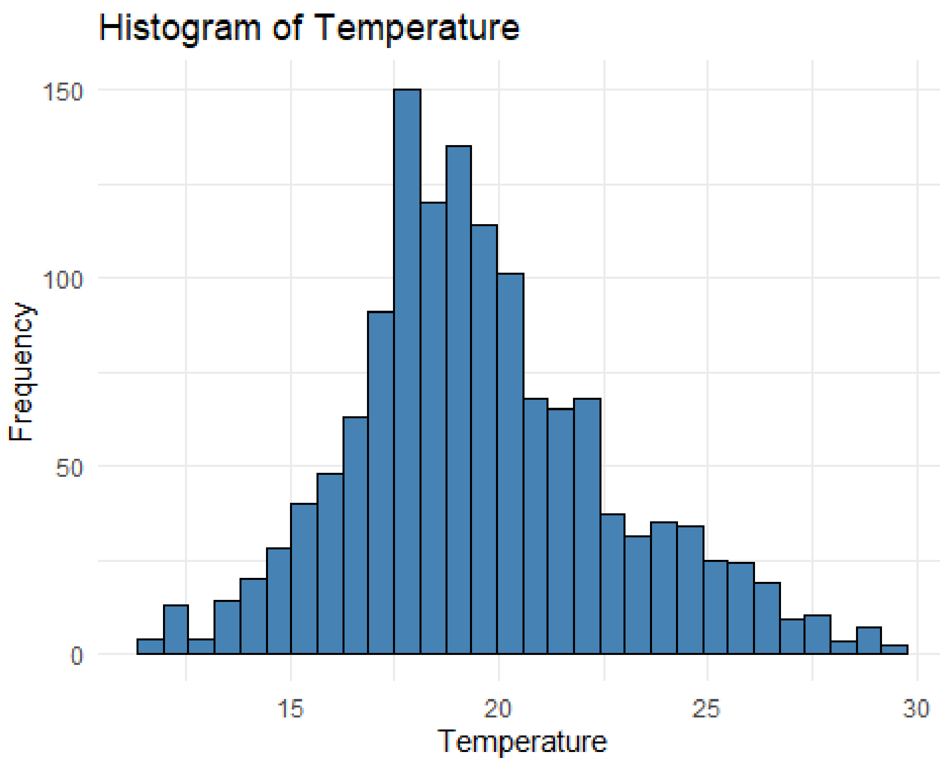
Create a histogram of atmospheric pressure faceted by 'province'

```
library(ggplot2)

# Create a histogram of atmospheric pressure faceted by 'province'
df <- climate_weather_df %>% filter(yr == 2010 & mo==08)
ggplot(df, aes(x = stp, fill = prov)) +
  geom_histogram(bins = 30, color = "black", alpha = 0.7) +

  facet_wrap(~prov, scales = "free") +

  # Customize labels and theme
  ggtitle("Histogram of atmospheric pressure by prov") +
  xlab("Atmospheric Pressure") +
  ylab("Frequency") +
  theme_minimal()
```



We create a histrogram of temperature which is a representation of frequency plot

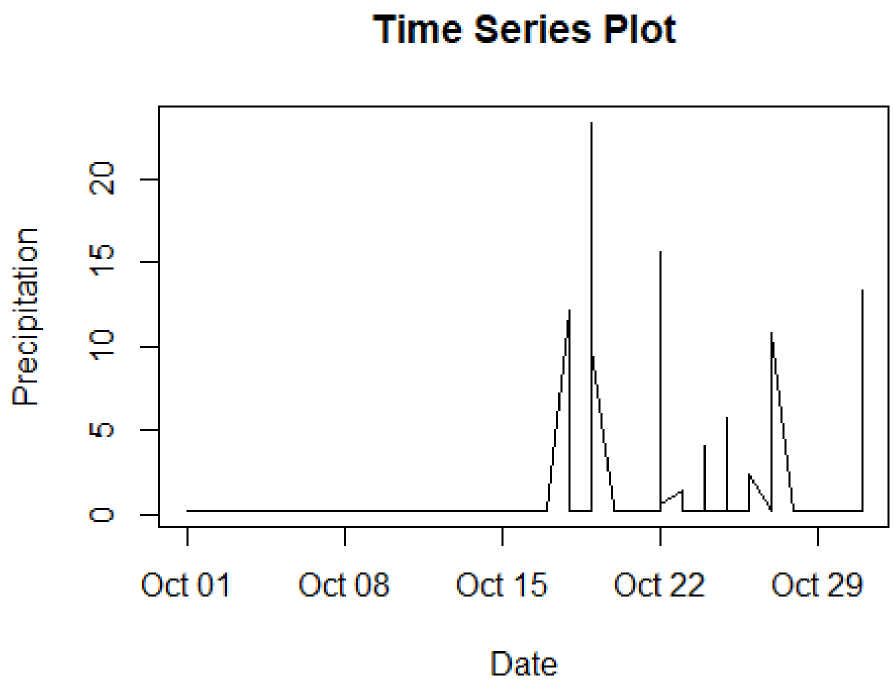
```
# Load ggplot2 package if already installed
library(ggplot2)

# Basic histogram with a border color
df <- climate_weather_df %>% filter(yr == 2005 & mo == 12)
ggplot(df, aes(x = temp)) +
  geom_histogram(color = "black", fill = "steelblue") +
  labs(x = "Temperature", y = "Frequency") +
  ggtitle("Histogram of Temperature") +
  theme_minimal()

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



We load xts and tsbox to create a time series plot() of different variables

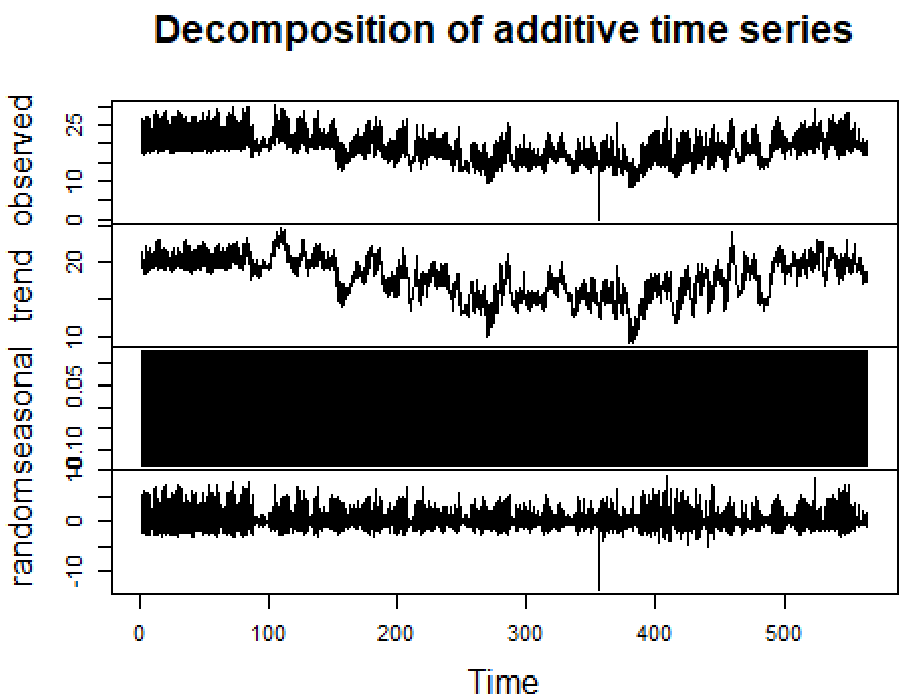
```
library(xts)

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric

##
## ######################### Warning from 'xts' package
##########################
## #
#
## # The dplyr lag() function breaks how base R's lag() function is supposed
to  #
## # work, which breaks lag(my_xts). Calls to lag(my_xts) that you type or
#
## # source() into this session won't work correctly.
#
## #
#
```

```
## # Use stats::lag() to make sure you're not using dplyr::lag(), or you can
add #
## # conflictRules('dplyr', exclude = 'lag') to your .Rprofile to stop
#
## # dplyr from breaking base R's lag() function.
#
## #
#
## # Code in packages is not affected. It's protected by R's namespace
mechanism #
## # Set `options(xts.warn_dplyr_breaks_lag = FALSE)` to suppress this
warning.  #
## #
#
##
###############################################################################
##

##
## Attaching package: 'xts'

## The following objects are masked from 'package:dplyr':
##
##     first, last

library(tsbox)

# Convert 'date' to Date class
climate_weather_df$date <- as.Date(climate_weather_df$date)

# Plot time series data
df <- climate_weather_df %>% filter(yr==2007 & mo==10 & wsnm=="BARBACENA")
plot(df$date, df$prcp, type = "l", xlab = "Date", ylab = "Precipitation",
main = "Time Series Plot")
```



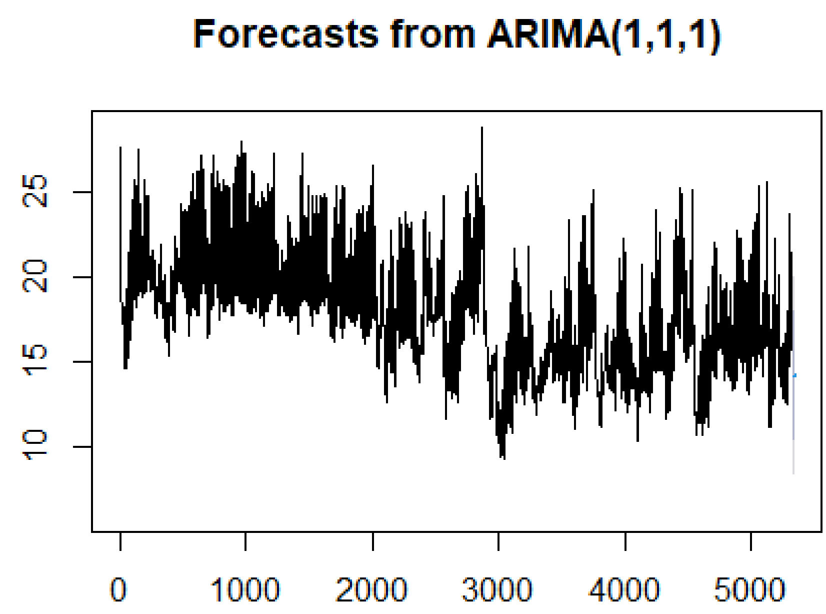
We create a Time series decomposition for daily maximum temperature

```
# Decompose time series
df <- climate_weather_df %>% filter(yr==2010 & wsnm=="SANTA TERESA")
decomposition <- decompose(ts(df$tmax, frequency = 12))  # Assuming monthly
data

# Plot decomposition
plot(decomposition)
```



We create a forecast for minimum temperature from ARIMA model

```
library(forecast)

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo

# Fit an ARIMA model
df <- climate_weather_df %>% filter(yr==2016 & wsnm=="SANTA TERESA")
arima_model <- arima(df$tmin, order = c(1,1,1))  # Example ARIMA(1,1,1) model

# Forecast future values
forecast_values <- forecast(arima_model)

# Plot forecast
plot(forecast_values)
```




```
Modelling with modelling verification
```

Select relevant features and target variable using correlation matrix

```r
# Load necessary libraries
library(caret)

## Loading required package: lattice

##
## Attaching package: 'caret'

## The following object is masked from 'package:purrr':
##
##     lift

# Select relevant features and target variable
features <- c('prcp','elvt', 'lat', 'lon', 'yr', 'mo', 'da', 'hr',
              'prcp', 'stp', 'smax', 'smin', 'gbrd', 'temp', 'tmax',
              'dmax', 'tmin', 'dmin', 'hmdy', 'hmax', 'hmin', 'wdsp', 'wdct',
'gust')
# Subset the data with selected features
climate_subset <- climate_weather_df[, c(features)]

# Compute correlation matrix
correlation_matrix <- cor(climate_subset)
```

```r
#Filter features based on correlation threshold (e.g., 0.5)
cor_threshold <- 0.5
highly_correlated <- findCorrelation(correlation_matrix, cutoff =
cor_threshold)
# Remove highly correlated features
selected_features <- features[highly_correlated]
selected_features

##  [1] "temp" "tmin" "tmax" "dmax" "dmin" "elvt" "smin" "stp"  "lon"  "hmin"
## [11] "hmdy" "gust" "prcp"
```

Load necessary libraries to build a decision tree and model prediction

```r
# Load necessary libraries
library(rpart)  # For decision tree
library(caret)  # For train/test split
library(rpart.plot)  # For plotting decision tree
```

Select relevant features and target variable

```r
features <- c( "temp", "tmin" ,"tmax" ,"dmax", "dmin", "elvt" ,"smin", "stp"
, "lon" , "hmin" ,"hmdy", "gust")
target <- "prcp"
```

Subset the data with selected features and target variable and split into train and test dataset

```r
climate_subset <- climate_weather_df[, c(features, target)]

# Train/test split
set.seed(123)  # for reproducibility
train_index <- createDataPartition(climate_subset$prcp, p = 0.7, list =
FALSE)
train_data <- climate_subset[train_index, ]
test_data <- climate_subset[-train_index, ]
```

Train decision tree model using rpart package

```r
model <- rpart(formula=prcp ~
temp+tmin+tmax+dmax+dmin+elvt+smin+stp+lon+hmin+hmdy+gust, data =
train_data,method="class")
```

Make predictions on train data and assess prediction accuracy on train data

```r
predictions <- predict(model,train_data,type="class")
```

```r
# Evaluate model
conMat <- confusionMatrix(as.factor(predictions), as.factor(train_data$prcp))
prediction_accuracy <- conMat$overall["Accuracy"]
prediction_accuracy

##  Accuracy
## 0.9286098
```

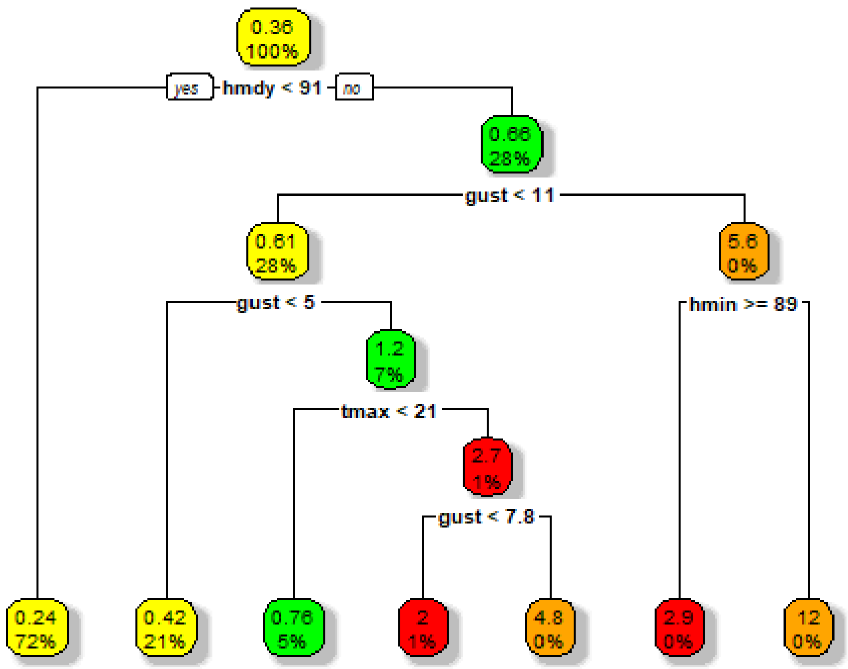
Make predictions on test dataset and evaluate model to predict accuracy

```r
model <- rpart(formula=prcp ~
temp+tmin+tmax+dmax+dmin+elvt+smin+stp+lon+hmin+hmdy+gust, data =
test_data,method="class")
predictions <- predict(model,test_data,type="class")
# Evaluate model
conMat <- confusionMatrix(as.factor(predictions), as.factor(test_data$prcp))

predict_test_accuracy <- format(conMat$overall["Accuracy"],digit=4)
predict_test_accuracy

## Accuracy
## "0.9287"
```

Display and plot the decision tree

```r
# Train decision tree model
model <- rpart(formula = prcp ~ ., data = train_data)
# Display the decision tree
print(model)

## n= 591188
##
## node), split, n, deviance, yval
##       * denotes terminal node
##
##  1) root 591188 961557.50  0.3572732
##    2) hmdy< 90.5 426411 114371.70  0.2409910 *
##    3) hmdy>=90.5 164777 826499.40  0.6581889
##      6) gust< 11.05 163063 640756.90  0.6058100
##       12) gust< 4.95 122860 177341.80  0.4208237 *
##       13) gust>=4.95 40203 446362.70  1.1711270
##         26) tmax< 20.85 31470 155588.40  0.7606419 *
##         27) tmax>=20.85 8733 266363.30  2.6503380
##           54) gust< 7.75 6701 135311.40  2.0102970 *
##           55) gust>=7.75 2032 119254.30  4.7610240 *
##      7) gust>=11.05 1714 142734.00  5.6413070
##       14) hmin>=88.5 1177  41844.31  2.9468140 *
##       15) hmin< 88.5 537  73614.50 11.5471100 *
```

```r
# Plot decision tree
rpart.plot(model, box.palette = c("yellow", "green","red","orange"), shadow.col="gray")
```



Let us do cross-validation on model.Setup model's train and valid dataset. Our cross validation settings are: 10 folds, and repeat 5 times, with Grid search the optimal parameter and Display details of the cross validation model

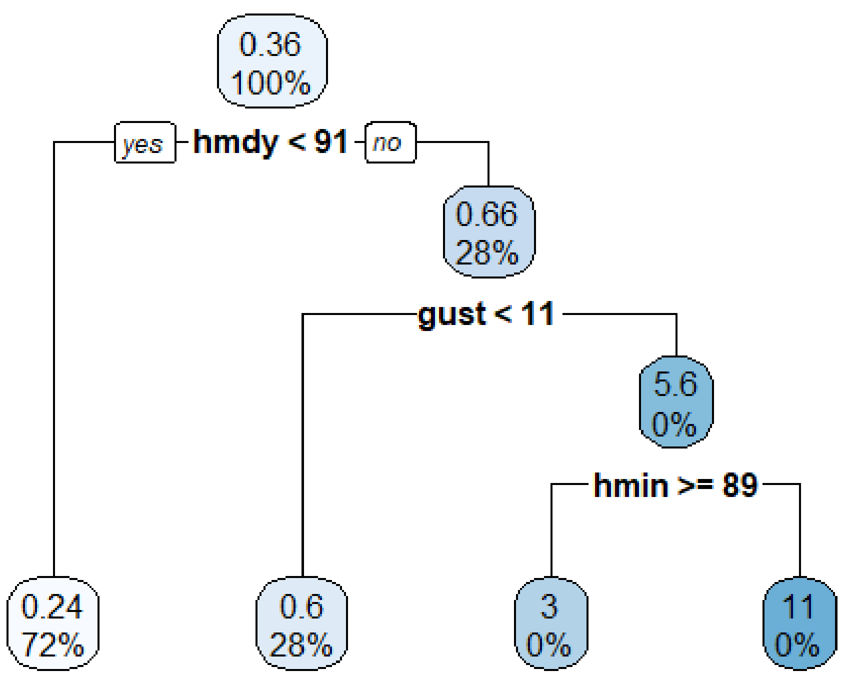
```r
#setup model's train and valid dataset
set.seed(1000)
samp <- sample(nrow(train_data), 0.8 * nrow(train_data))
trainData <- train_data[samp, ]
validData <- train_data[-samp, ]
# set random for reproduction
set.seed(3214)
# specify parameters for cross validation
control <- trainControl(method = "repeatedcv",
                        number = 10, # number of folds
                        repeats = 5, # repeat times
                        search = "grid")
tree_model_cv <- train(prcp ~
temp+tmin+tmax+dmax+dmin+elvt+smin+stp+lon+hmin+hmdy+gust, data =
trainData,method="rpart", trControl = control)
```

```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info =
trainInfo,
## : There were missing values in resampled performance measures.
```

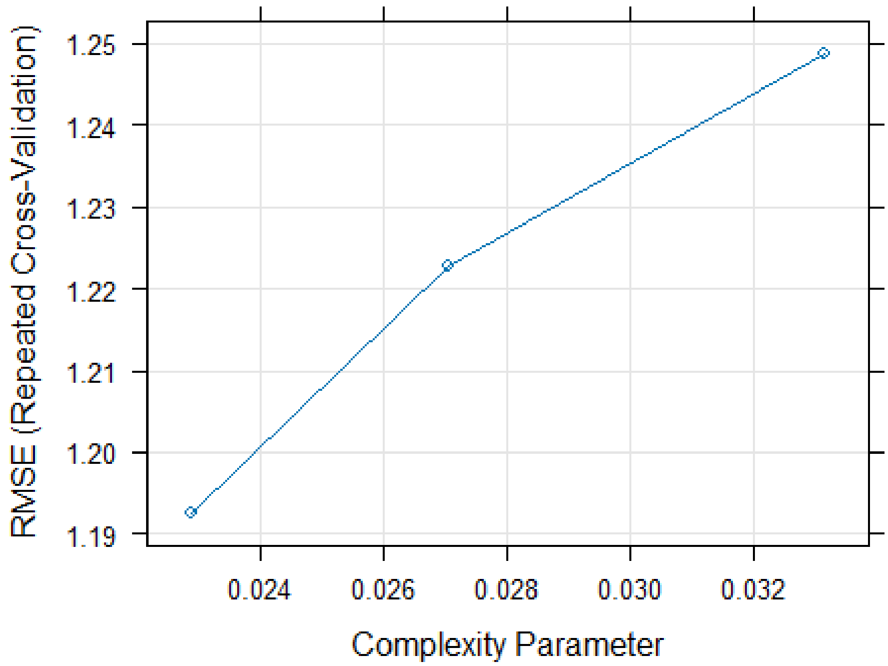
```r
print.train(tree_model_cv)
```

```
## CART
##
## 472950 samples
##     12 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold, repeated 5 times)
## Summary of sample sizes: 425654, 425655, 425655, 425654, 425655, 425656,
...
## Resampling results across tuning parameters:
##
##   cp          RMSE      Rsquared    MAE
##   0.02288329  1.192589  0.11674842  0.2585639
##   0.02705115  1.222813  0.07117864  0.2702686
##   0.03313536  1.248725  0.05483589  0.2830002
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was cp = 0.02288329.
```

```r
#Visualize cross validation tree
rpart.plot(tree_model_cv$finalModel)
```



```r
plot.train(tree_model_cv)
```



Let us record the model's accuracy on trainData, validData, and test dataset. Remember trainData and validData are randomly partitioned from the train dataset.

```r
tree_model_cv <- rpart(prcp ~
temp+tmin+tmax+dmax+dmin+elvt+smin+stp+lon+hmin+hmdy+gust, data =
trainData,method="class")
predict_train <-predict(tree_model_cv, trainData,type="class")
conMat <- confusionMatrix(as.factor(predict_train),
as.factor(trainData$prcp))
# prediction's Accuracy on the trainData
predict_train_accuracy <- format(conMat$overall["Accuracy"], digits=4)
paste("trainData Accuracy:", predict_train_accuracy)

## [1] "trainData Accuracy: 0.9289"

# prediction's Accuracy on the validData
predict_valid_accuracy <- format(conMat$overall["Accuracy"], digits=4)
paste("validData Accuracy:", predict_valid_accuracy)

## [1] "validData Accuracy: 0.9289"
```

It shows that the model construction has reached the best since the change of the tree structure does not increase the accuracy.

```r
# accumulate model's accuracy
name <- c( "Train Accu", "Valid Accu", "Test Accu")
Tree_model_CV_accuracy <- c(predict_train_accuracy, predict_valid_accuracy,
predict_test_accuracy)
names(Tree_model_CV_accuracy) <- name
Tree_model_CV_accuracy
```

```
## Train Accu Valid Accu  Test Accu
##   "0.9289"   "0.9289"   "0.9287"
```